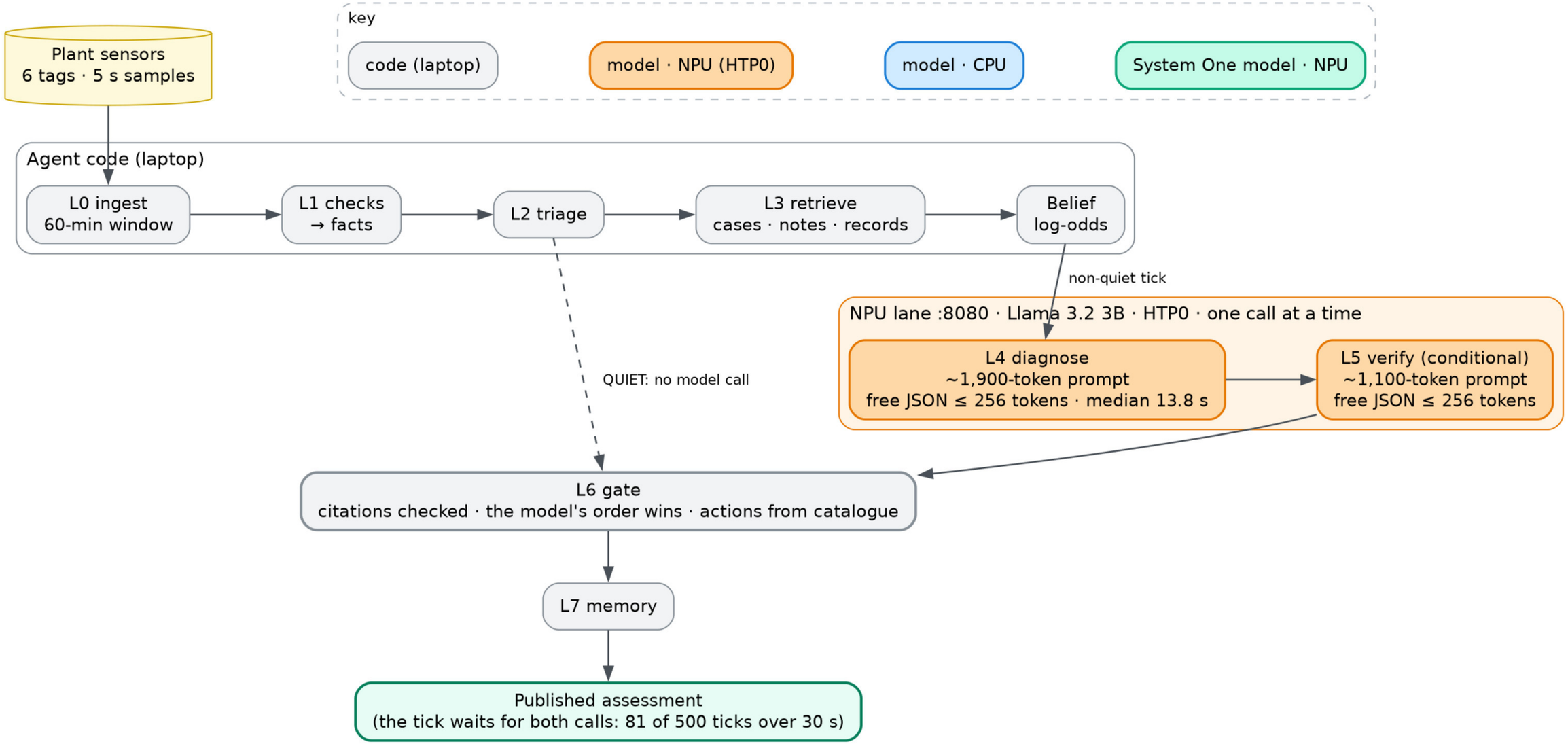
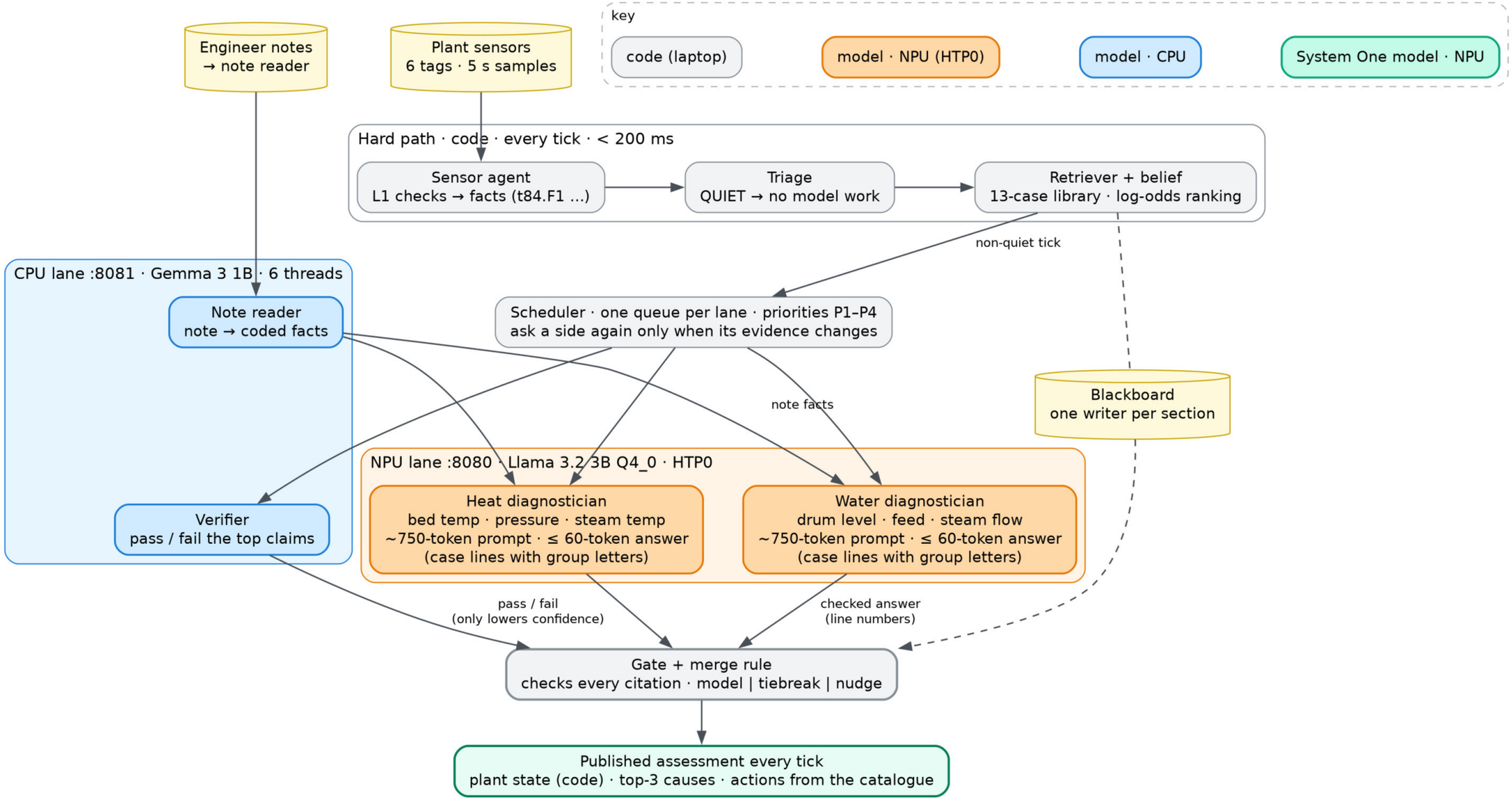
Single agent (baseline)

one long model call per tick, and the tick waits for it · Llama 3.2 3B Q4_0 on the NPU



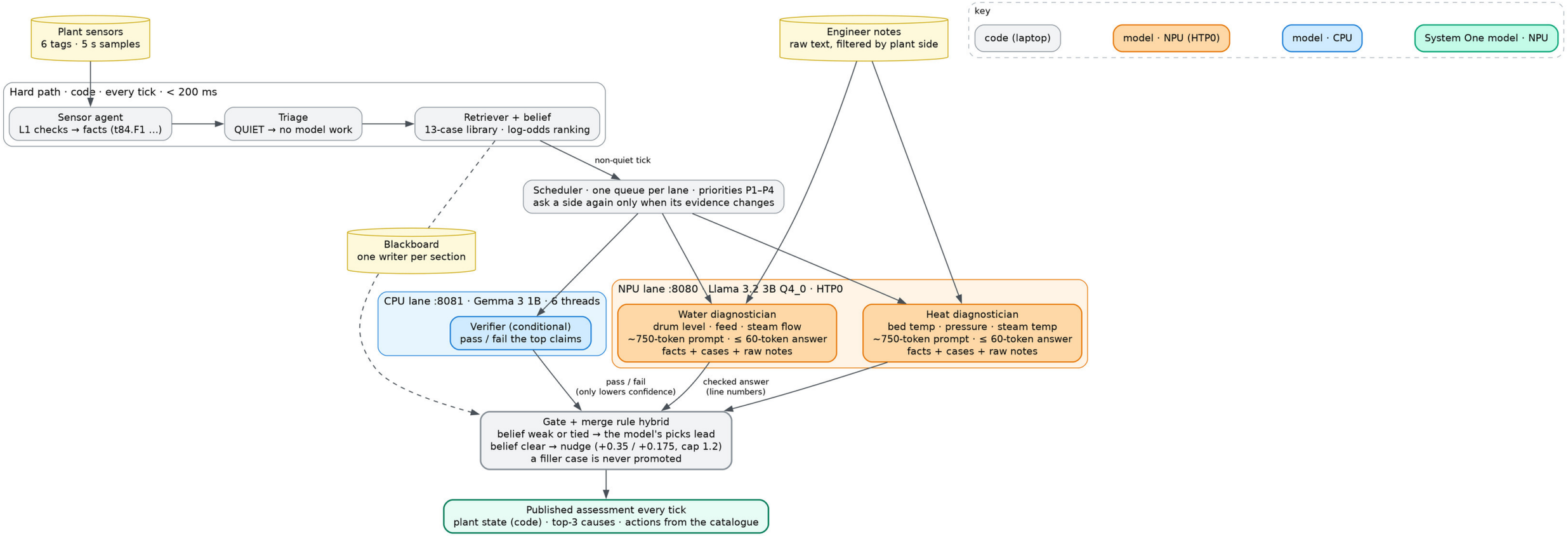
Multi-agent v2

small jobs on two lanes · grammar-constrained answers · case lines tagged with group letters · Gemma note reader · merge rule model / tiebreak / nudge



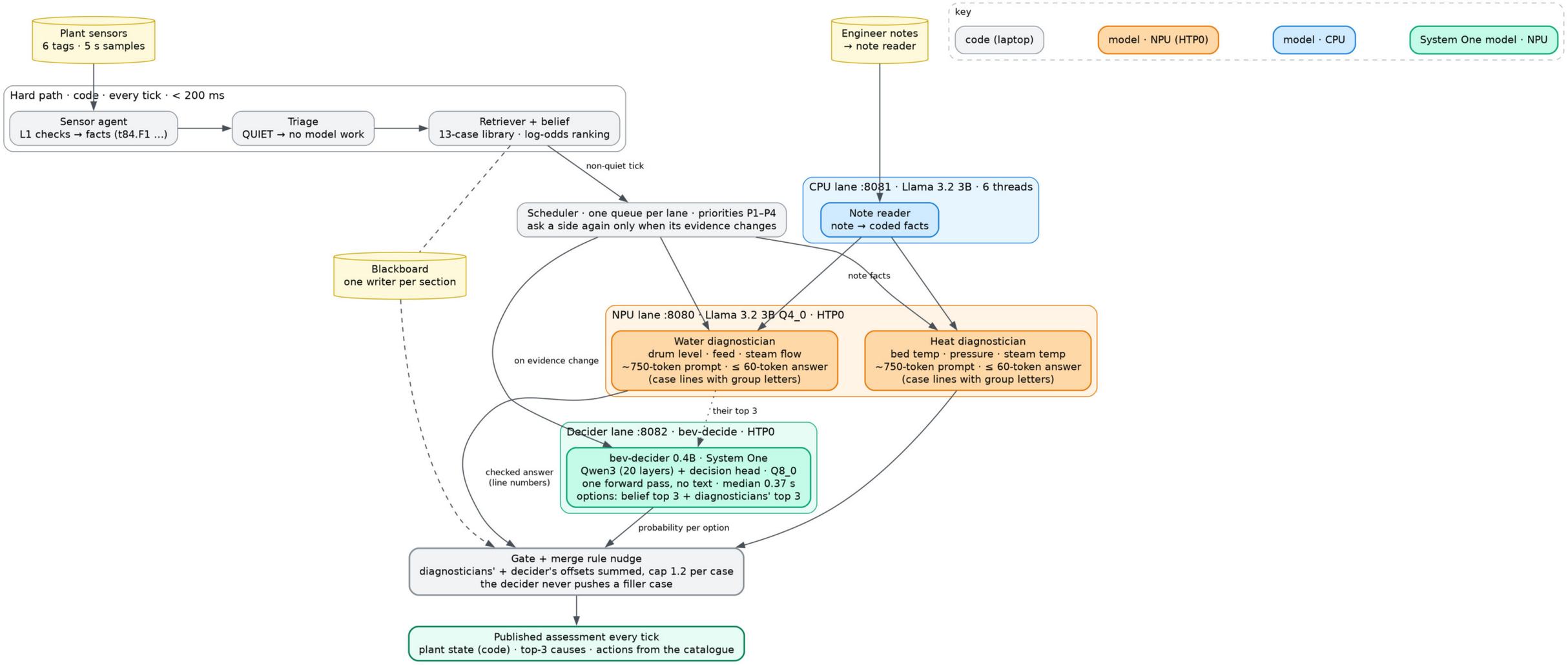
Multi-agent v3 (current best)

no group letters · raw note text straight to the diagnosticians · merge rule hybrid with the filler-case guard



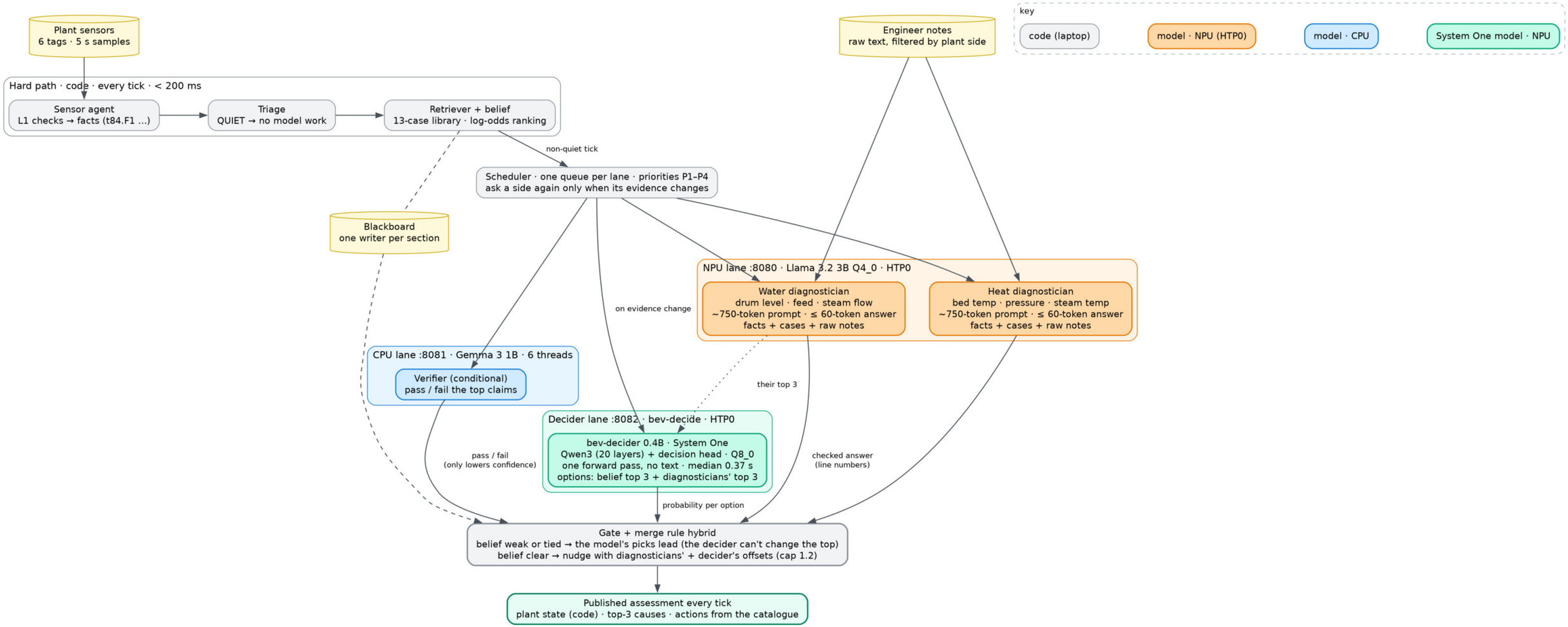
bev-decider-2

accuracy.yaml (v2 prompts with group letters, Llama 3B on both lanes, verifier off) + merge rule nudge + the System One decider



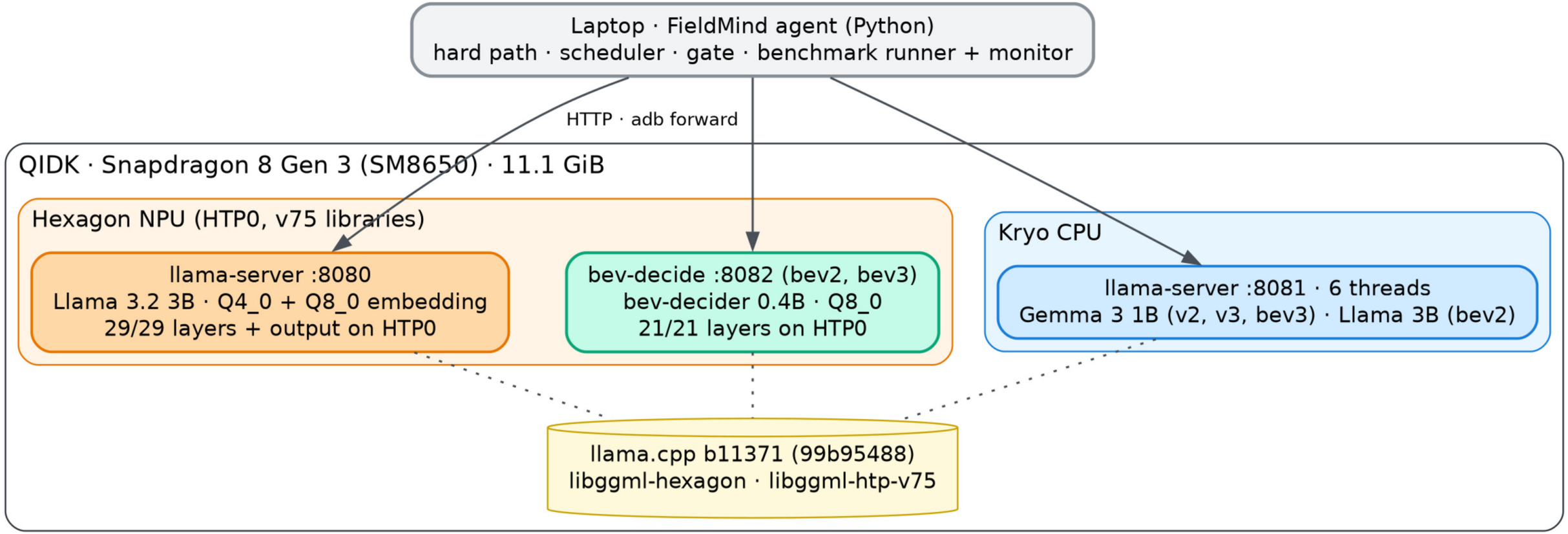
bev-decider-3 (v3 + System One decider)

exactly v3 plus the decider · three model servers on the board, two of them sharing the Hexagon NPU



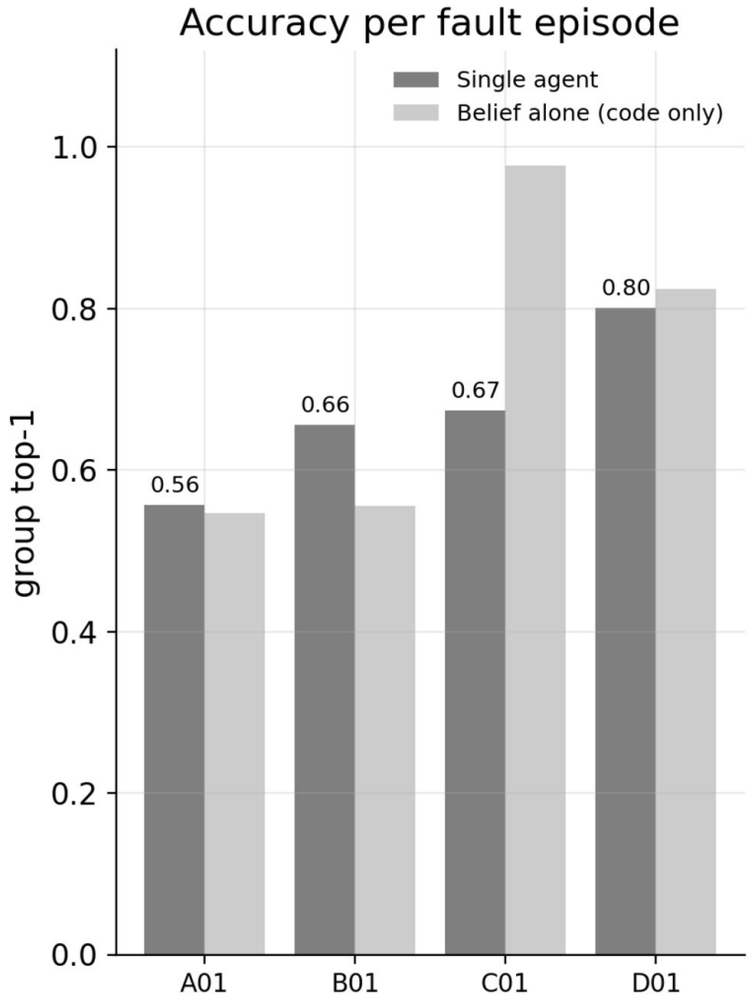
Deployment: laptop and QIDK board

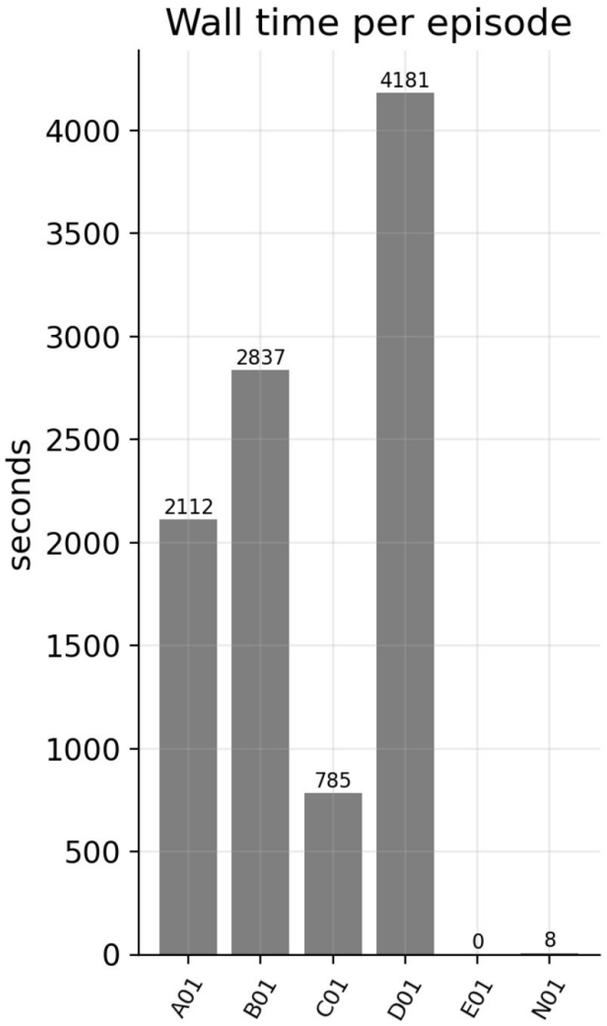
the agent runs on the laptop; every model call goes over USB (adb forward) to persistent servers on the board

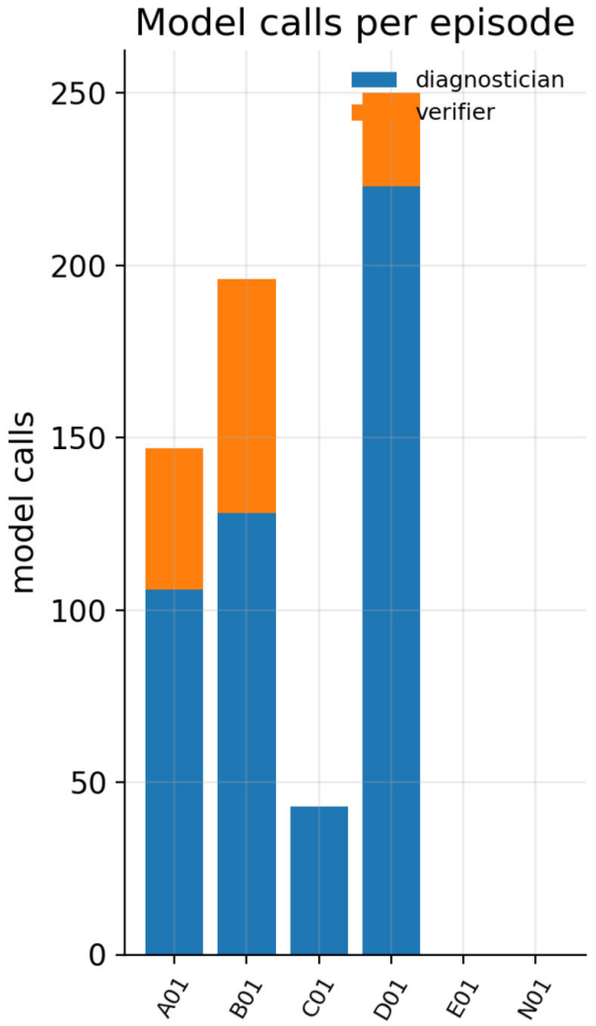


Benchmark: Single agent

L0-L7 pipeline, Llama 3.2 3B on the NPU, one big prompt, no grammar, model order wins
Mean group top-1 0.672 (belief 0.726) on 4 fault episodes · total wall 9,924 s · unusable answers 71 · ticks over 30 s 81



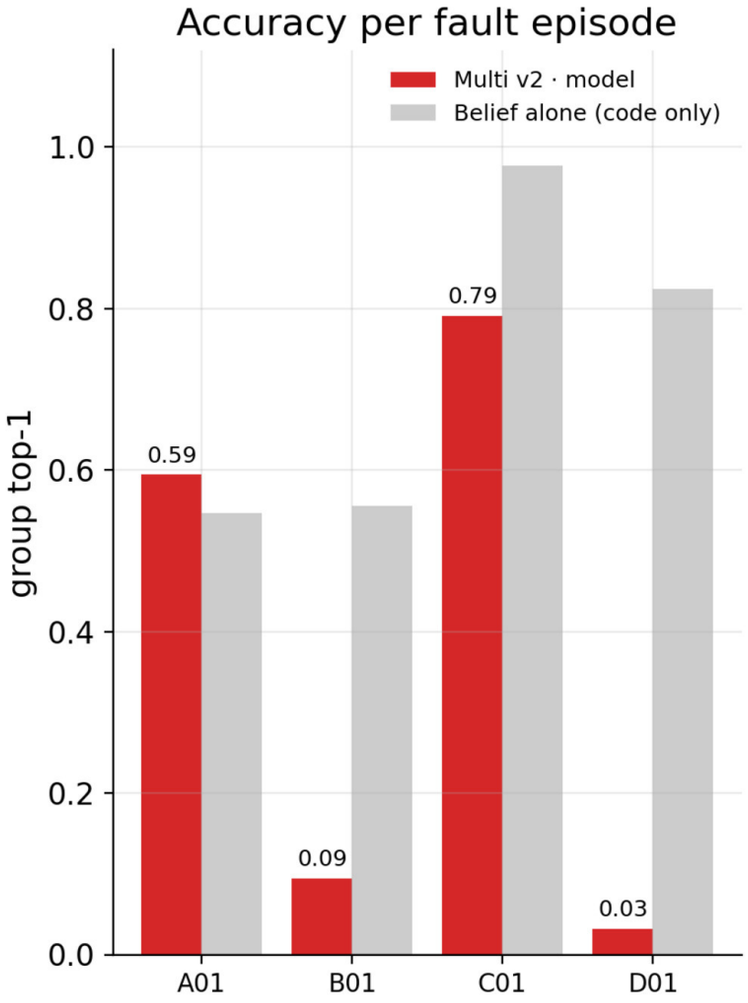


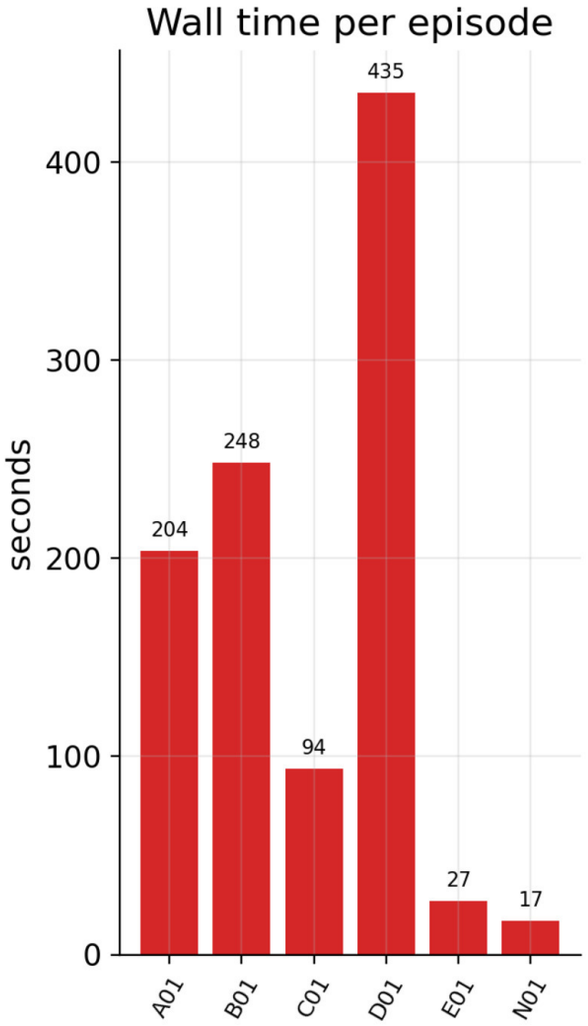


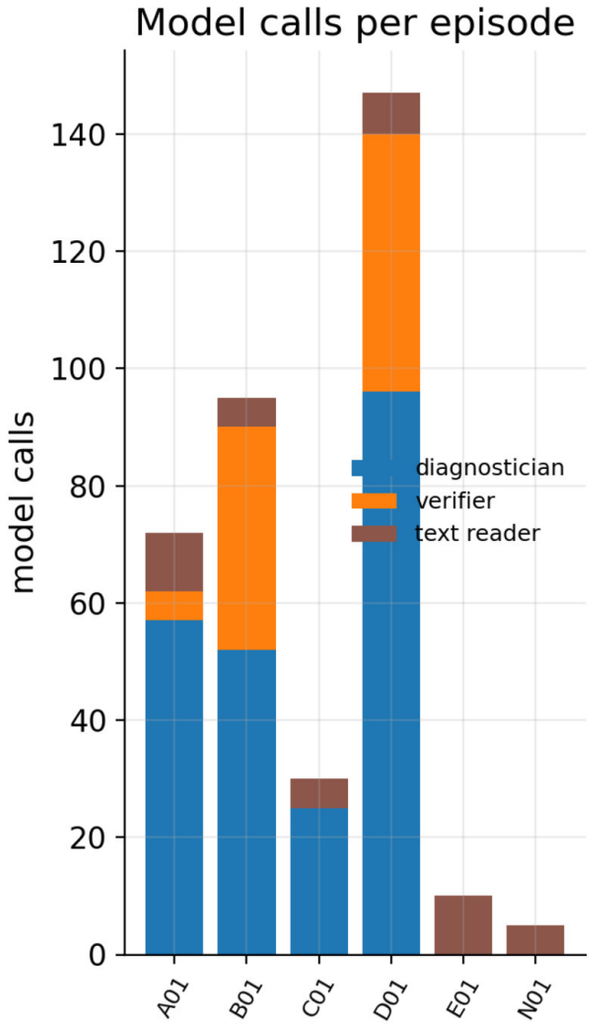
QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Benchmark: Multi v2 · model

multi-agent v2 prompts (group letters, Gemma note reader), merge rule model
Mean group top-1 0.378 (belief 0.726) on 4 fault episodes · total wall 1,024 s · unusable answers 0 · ticks over 30 s 0



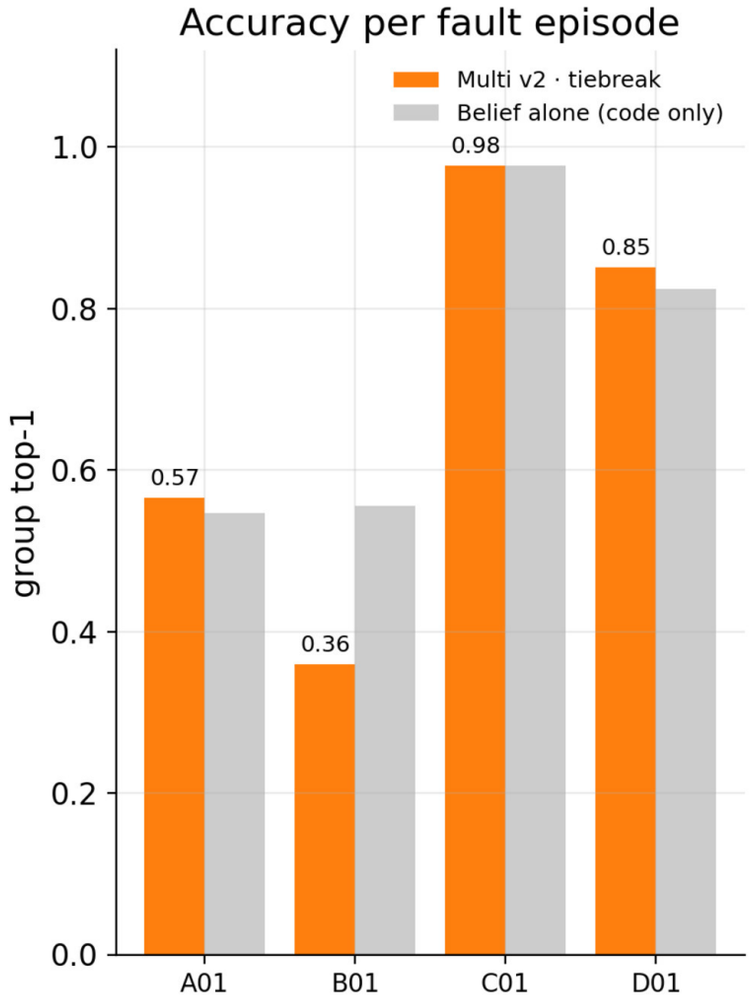


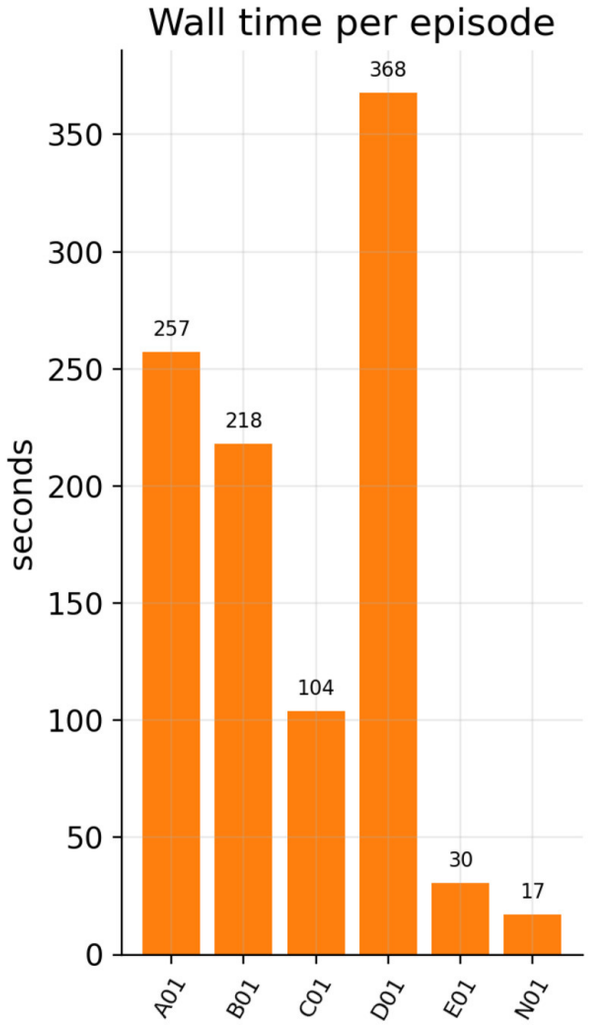


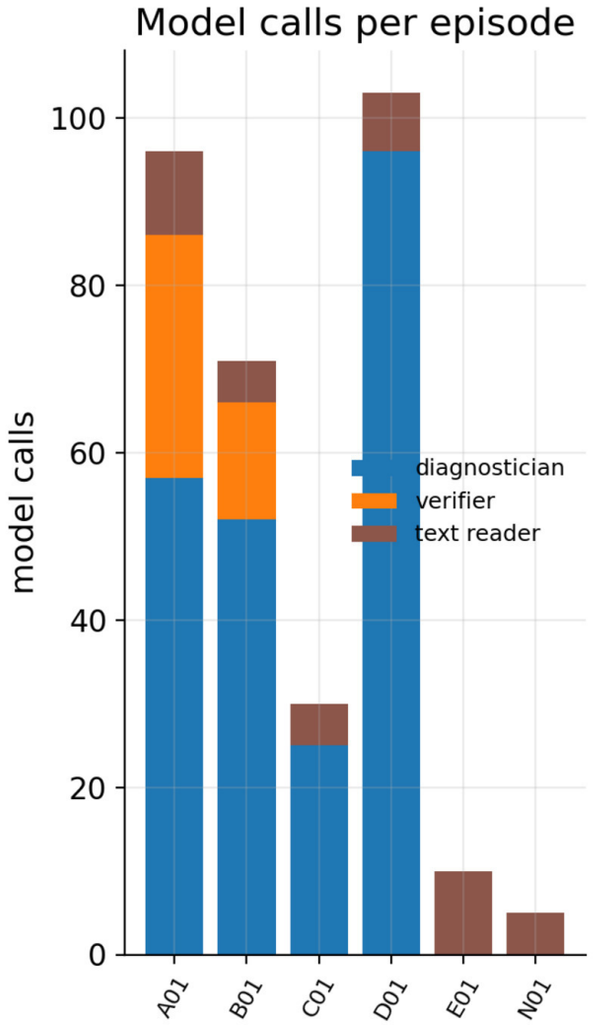
QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Benchmark: Multi v2 · tiebreak

multi-agent v2 prompts, merge rule tiebreak
Mean group top-1 0.688 (belief 0.726) on 4 fault episodes · total wall 994 s · unusable answers 0 · ticks over 30 s 0



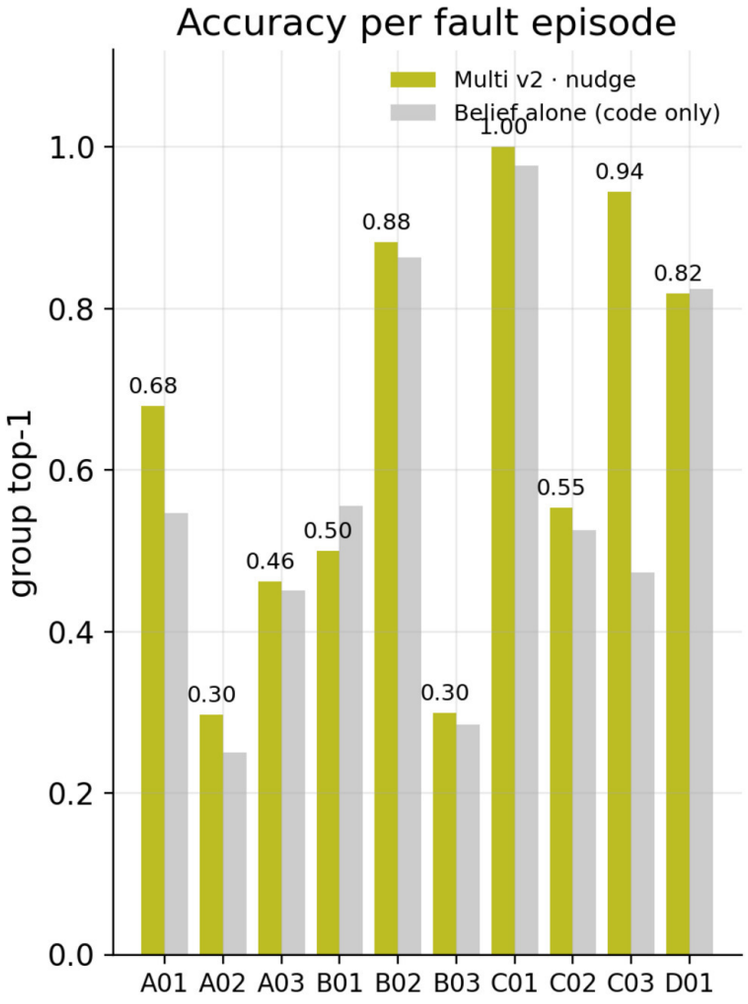


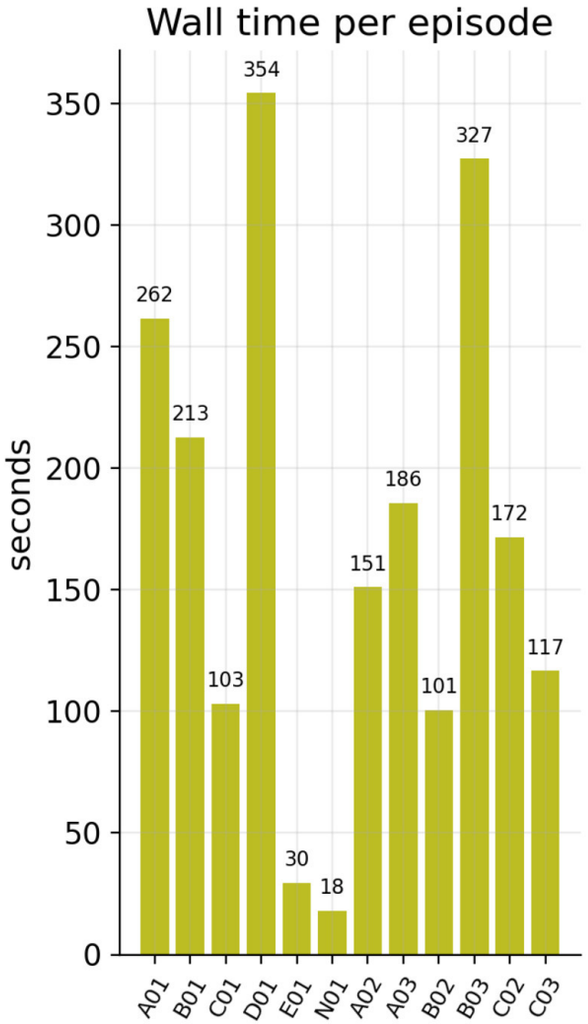


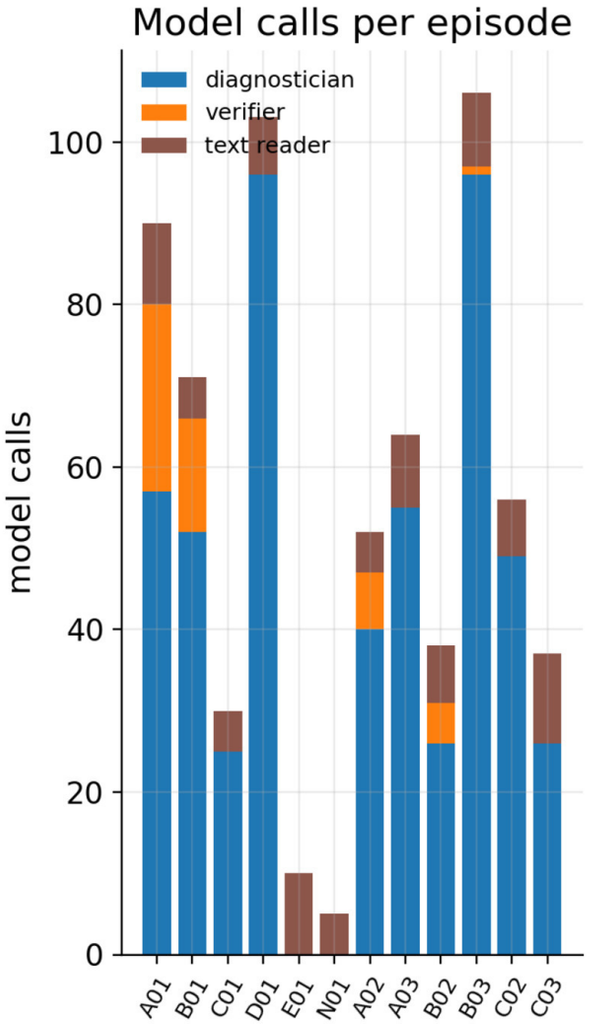
QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Benchmark: Multi v2 · nudge

multi-agent v2 prompts, merge rule nudge
Mean group top-1 0.644 (belief 0.575) on 10 fault episodes · total wall 2,032 s · unusable answers 0 · ticks over 30 s 0



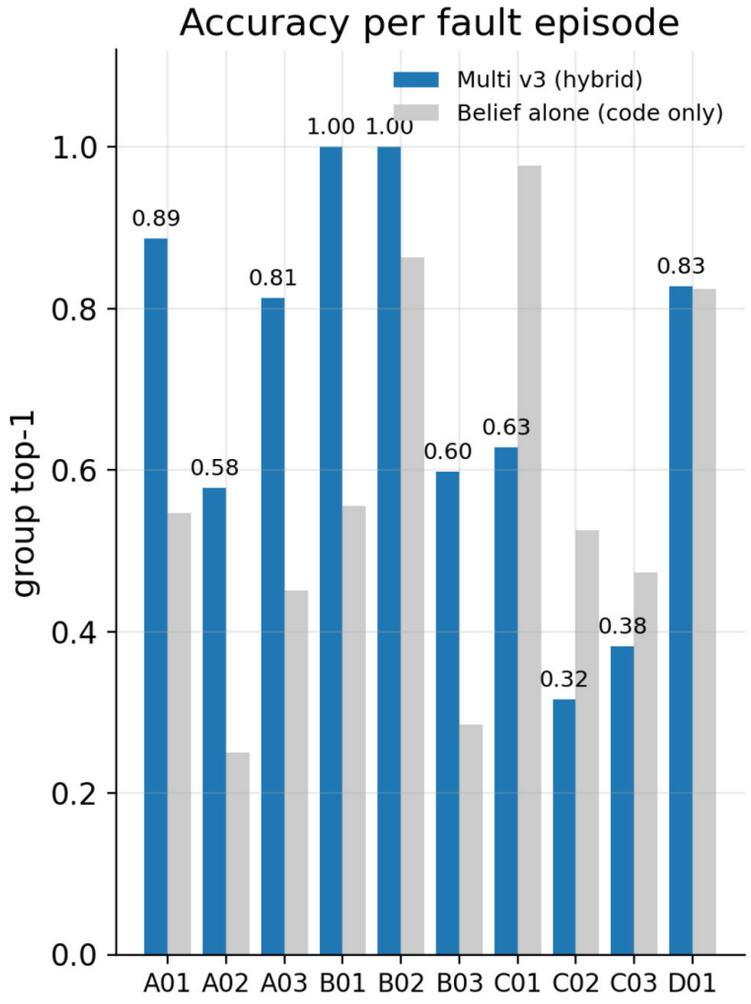


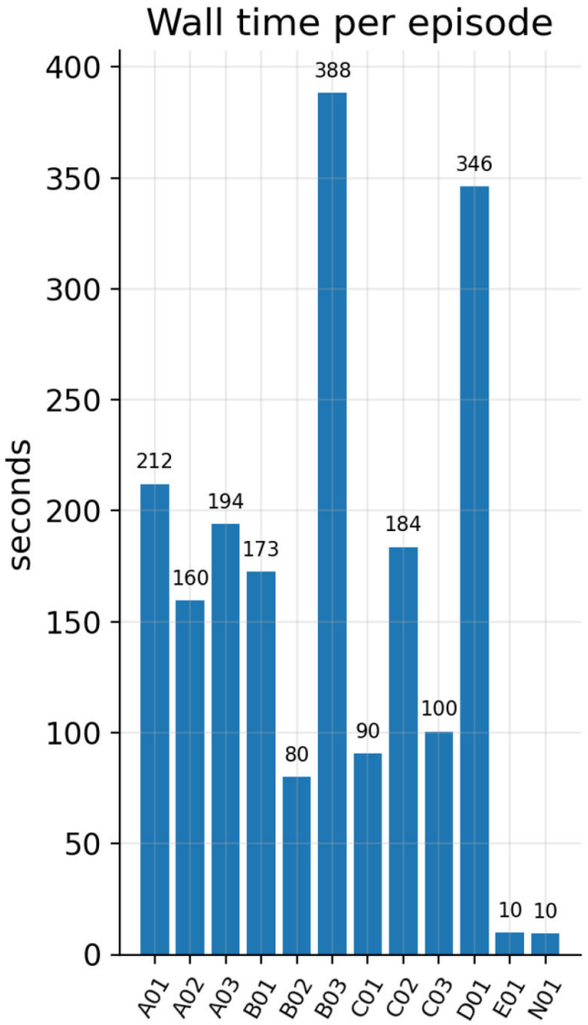


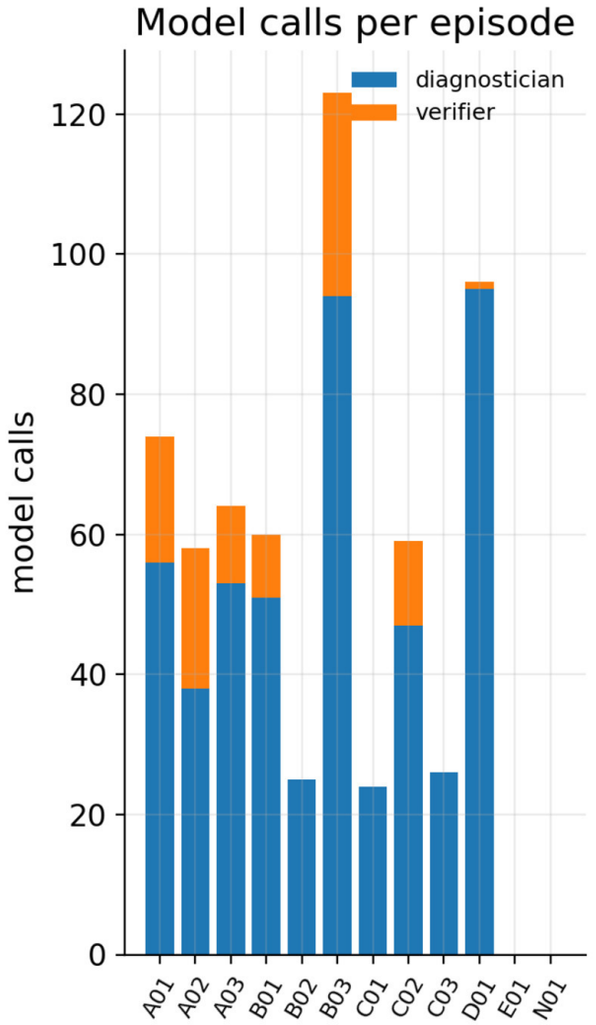
QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Benchmark: Multi v3 (hybrid)

multi-agent v3: no group letters, raw notes, merge rule hybrid + flat guard
Mean group top-1 0.703 (belief 0.575) on 10 fault episodes · total wall 1,947 s · unusable answers 0 · ticks over 30 s 0



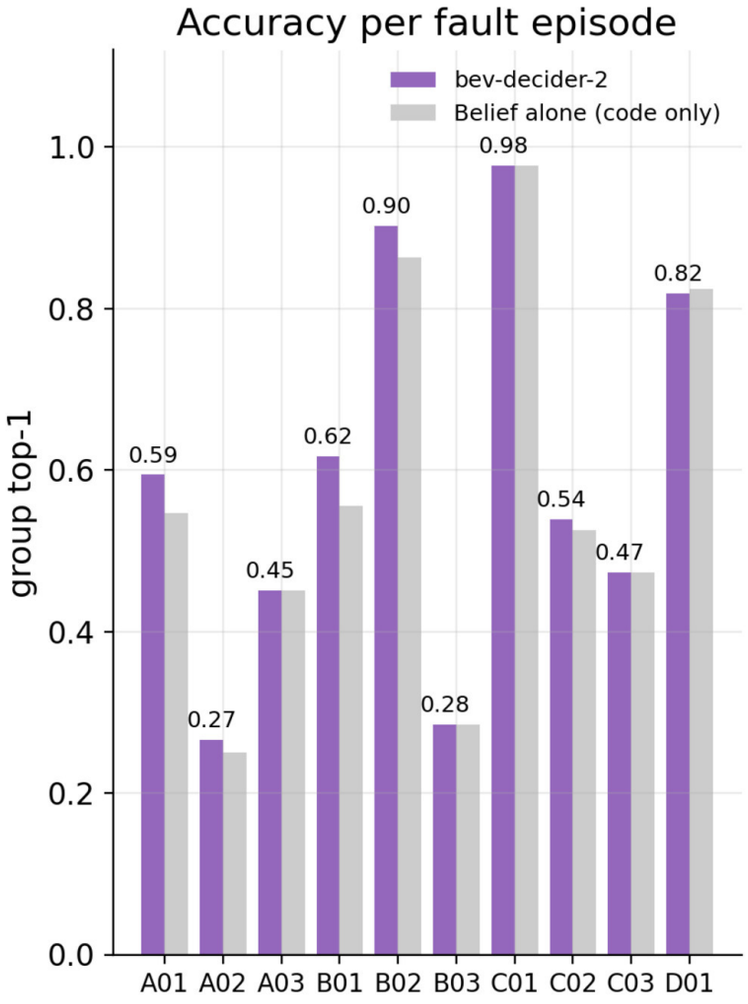


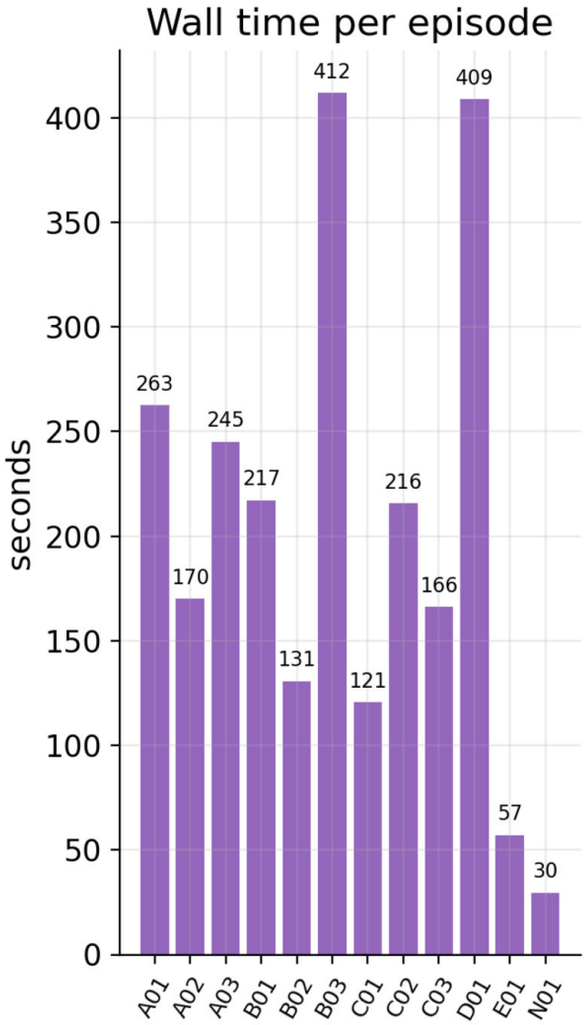


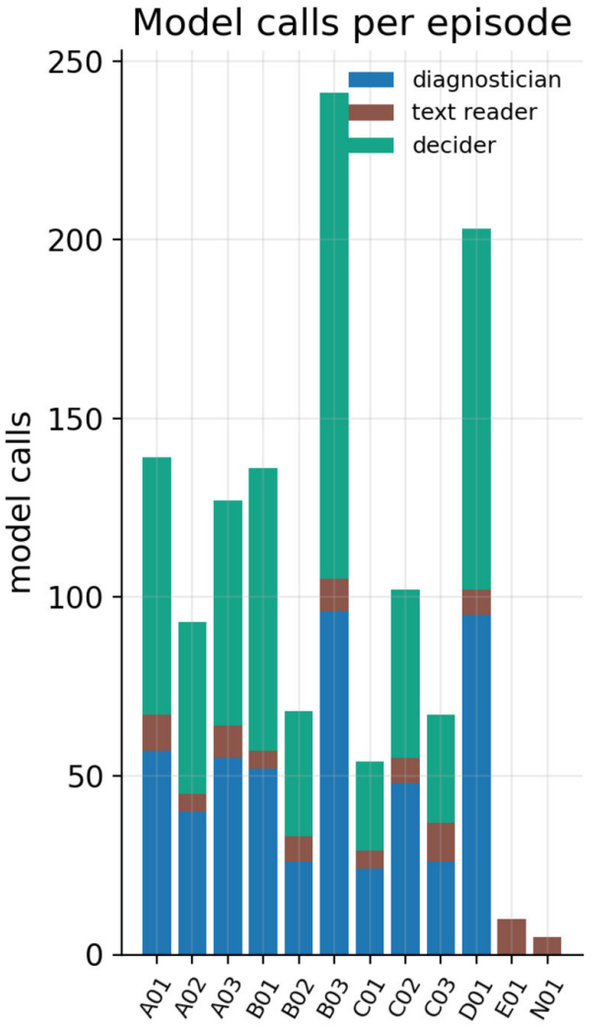
QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Benchmark: bev-decider-2

accuracy.yaml (v2 prompts, Llama 3B both lanes, verifier off) + nudge + bev-decider
Mean group top-1 0.592 (belief 0.575) on 10 fault episodes · total wall 2,435 s · unusable answers 0 · ticks over 30 s 0



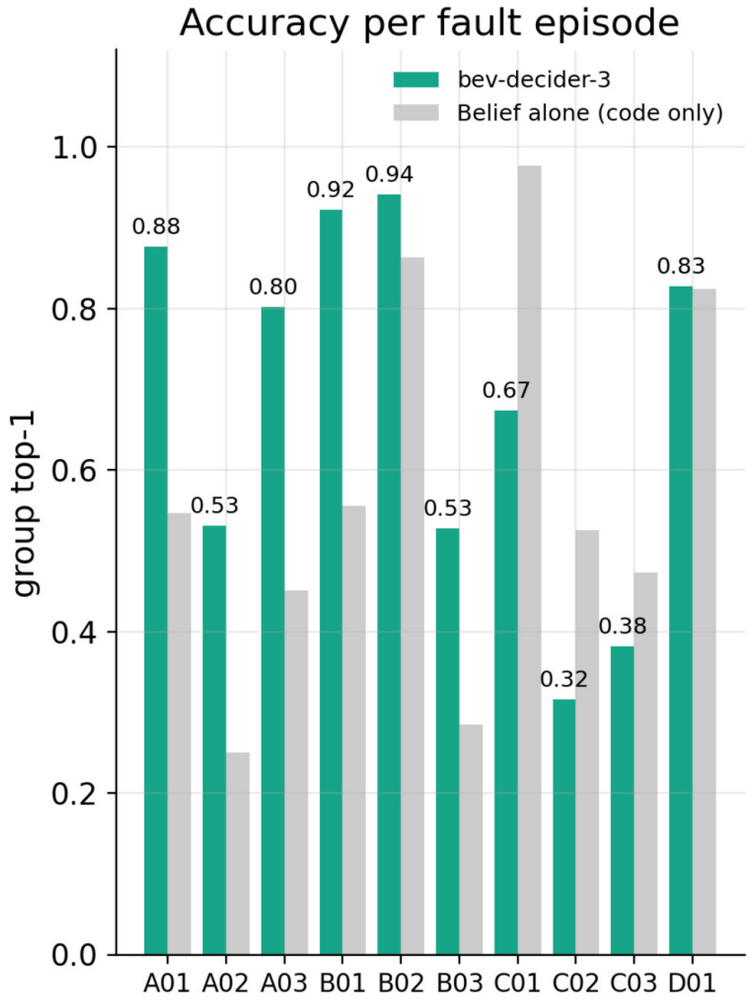


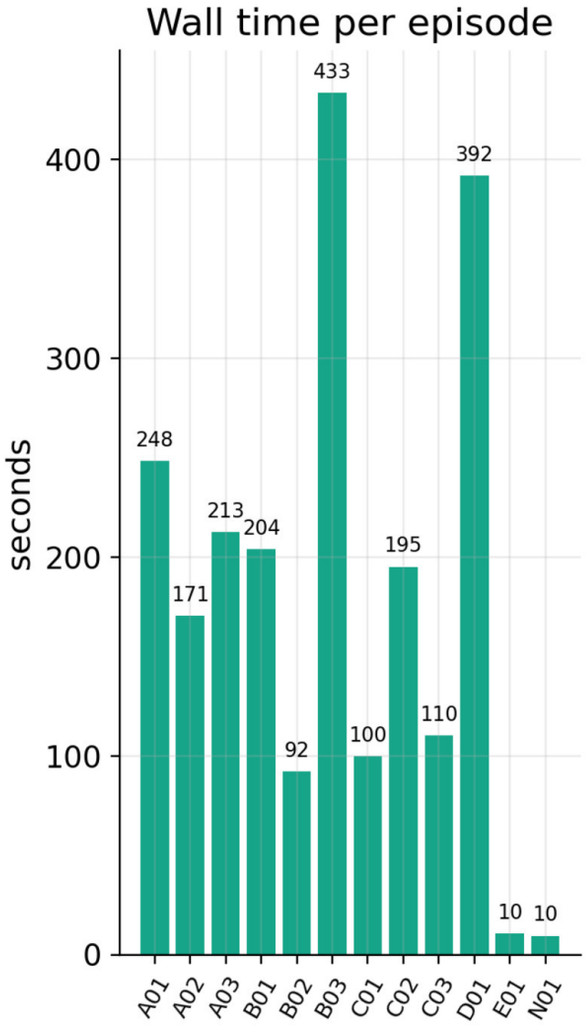


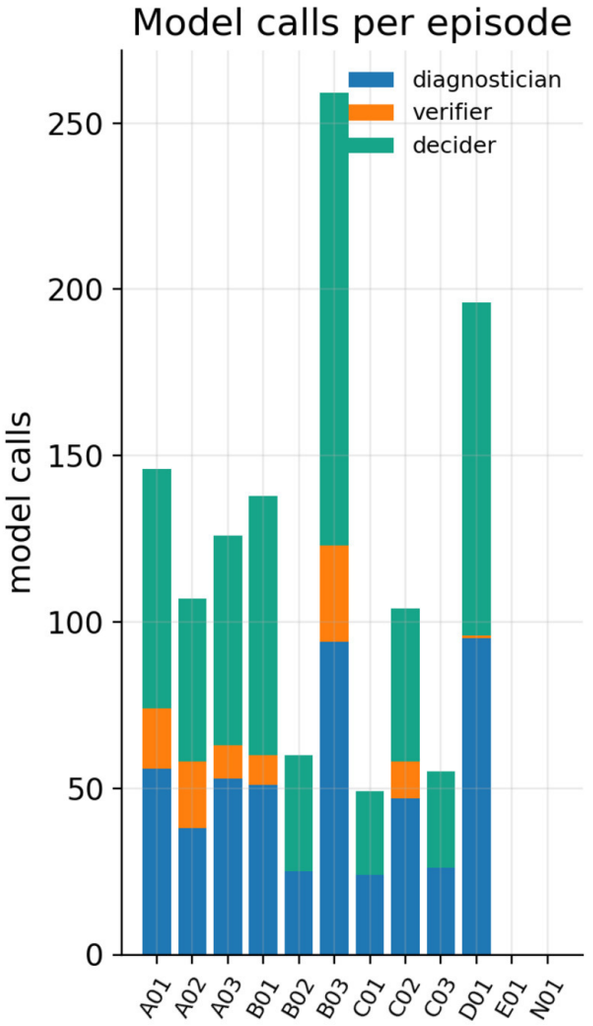
QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Benchmark: bev-decider-3

v3 + bev-decider (System One decider on the NPU, port 8082)
Mean group top-1 0.680 (belief 0.575) on 10 fault episodes · total wall 2,177 s · unusable answers 0 · ticks over 30 s 0



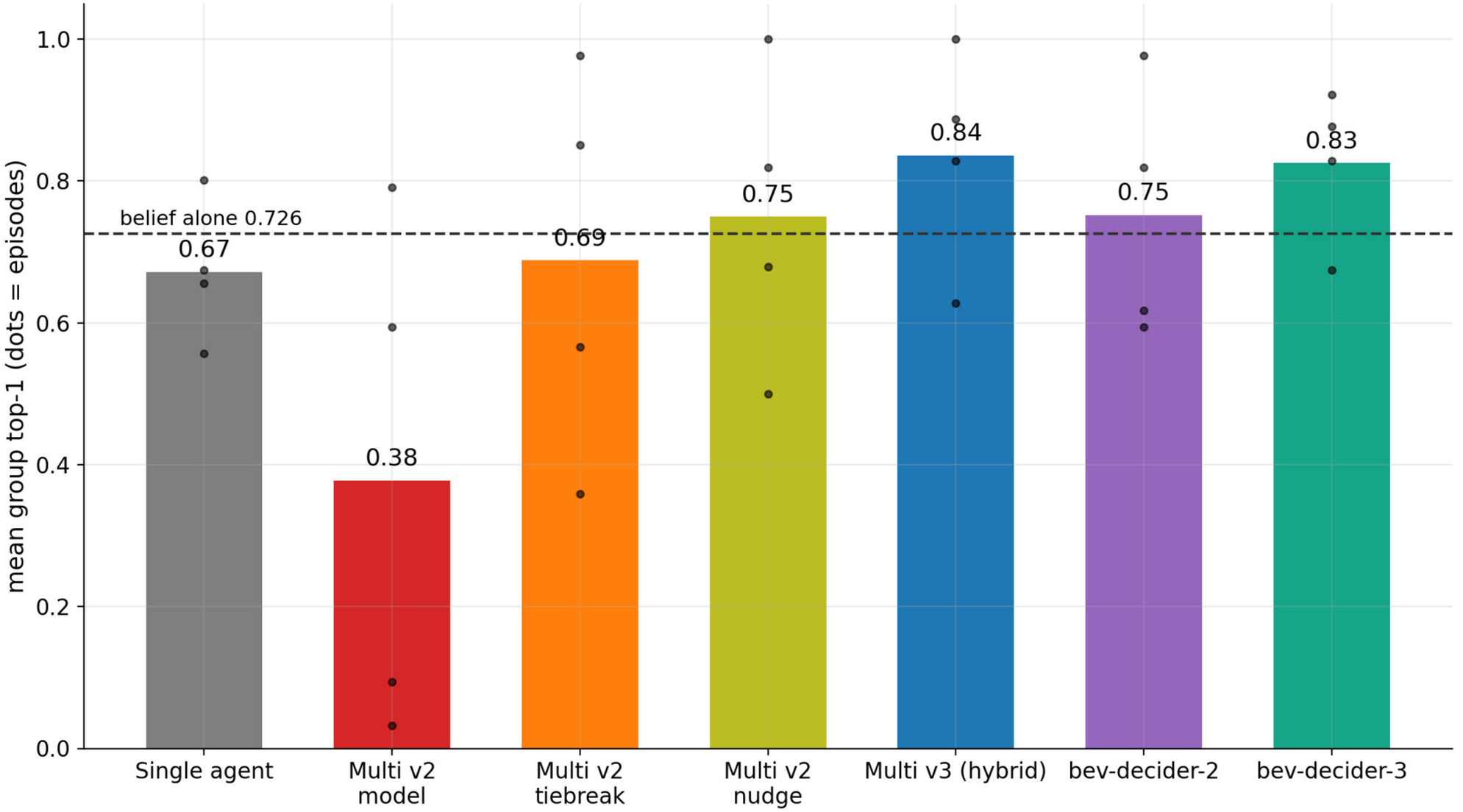




QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Accuracy: every configuration on the same four fault episodes

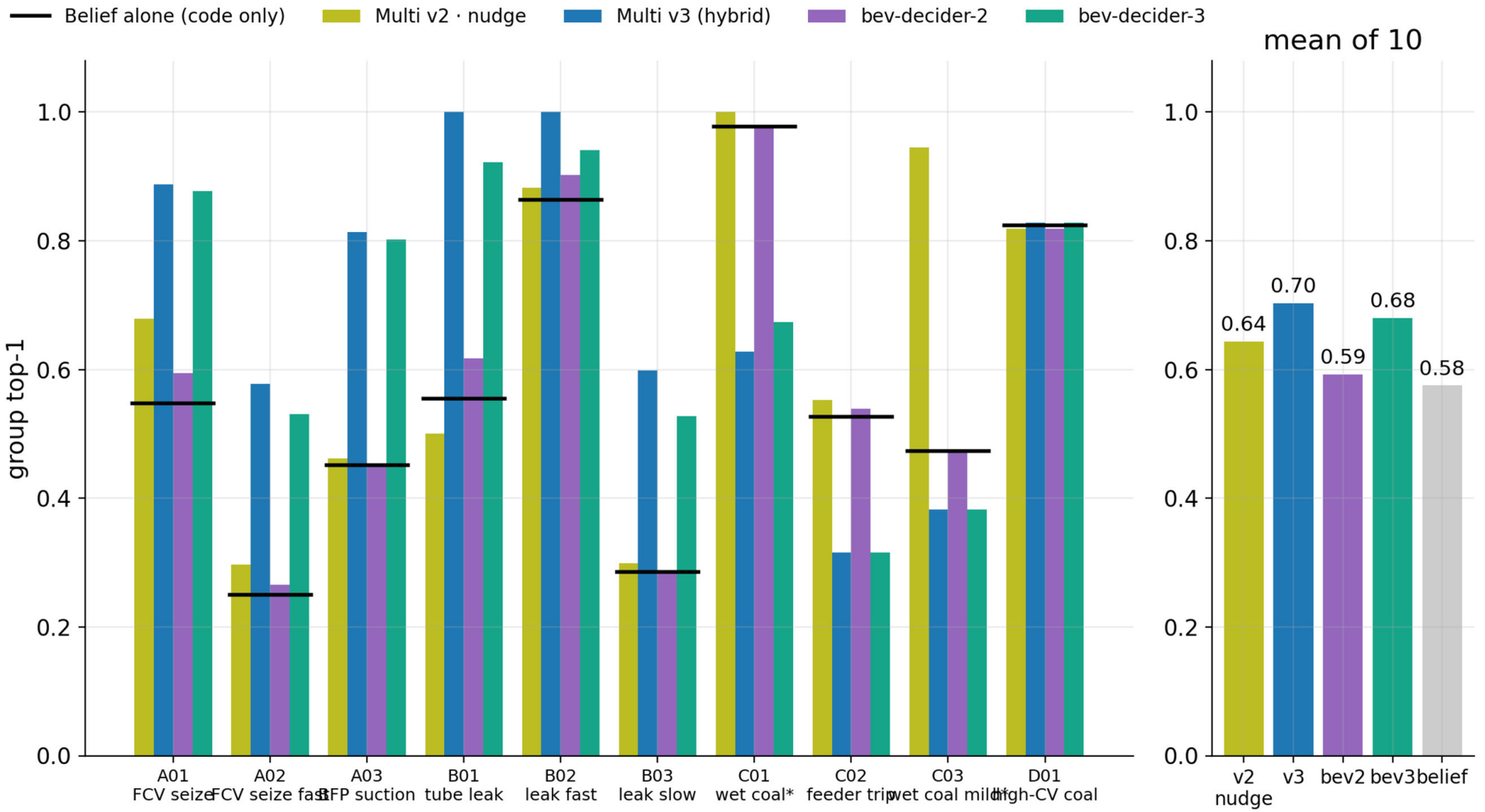
A01, B01, C01, D01 · bar = mean of the four episodes · dashed line = belief (code-only ranking)



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Accuracy on all ten fault episodes

* C01 and C03: the true cause (wet coal) is held out of the library; only its look-alike group can score



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Group top-1, every configuration × fault episode

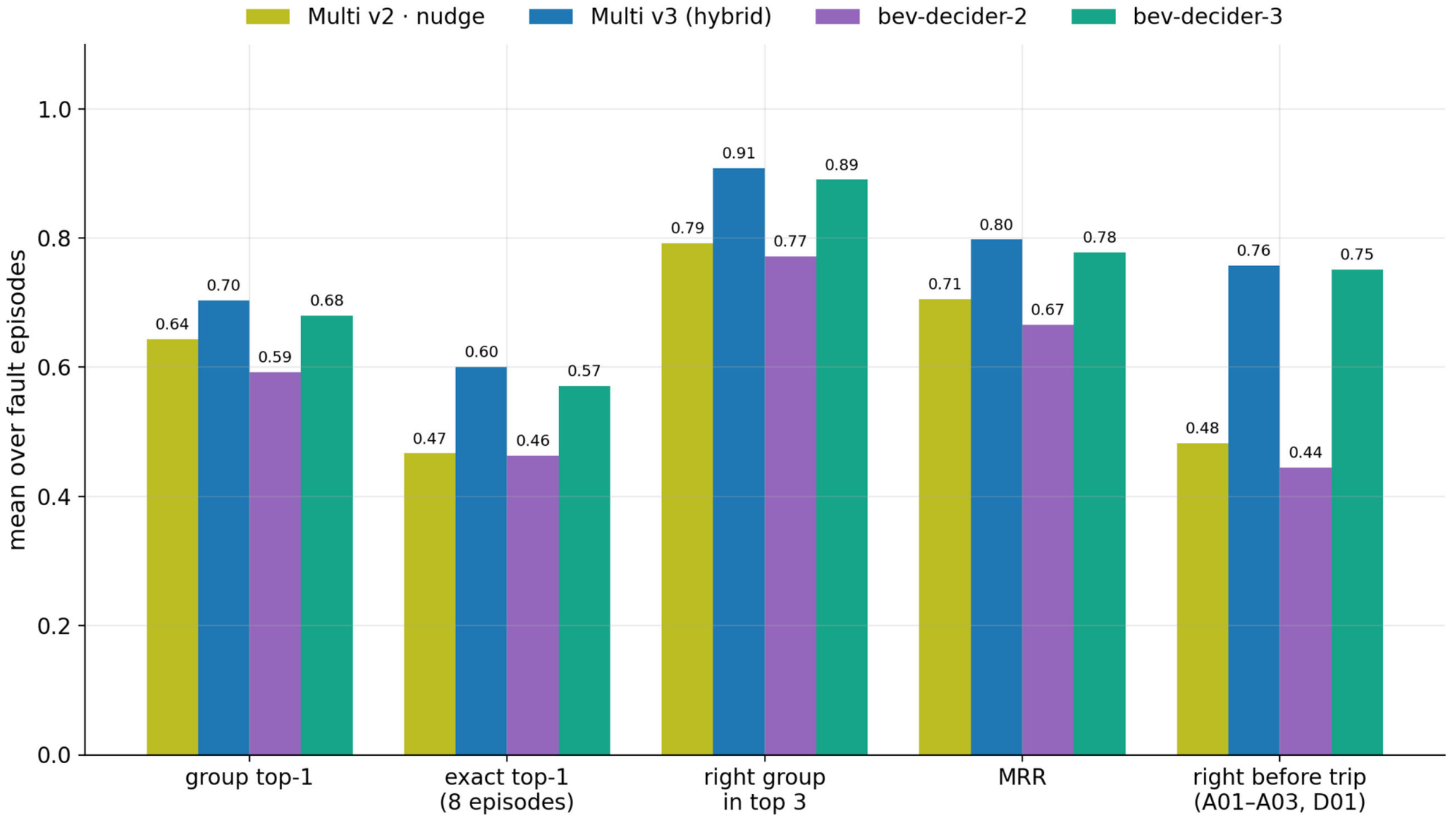
Belief row: the code-only ranking (the same in every run: computed by code)



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Diagnosis quality: five metrics, ten fault episodes

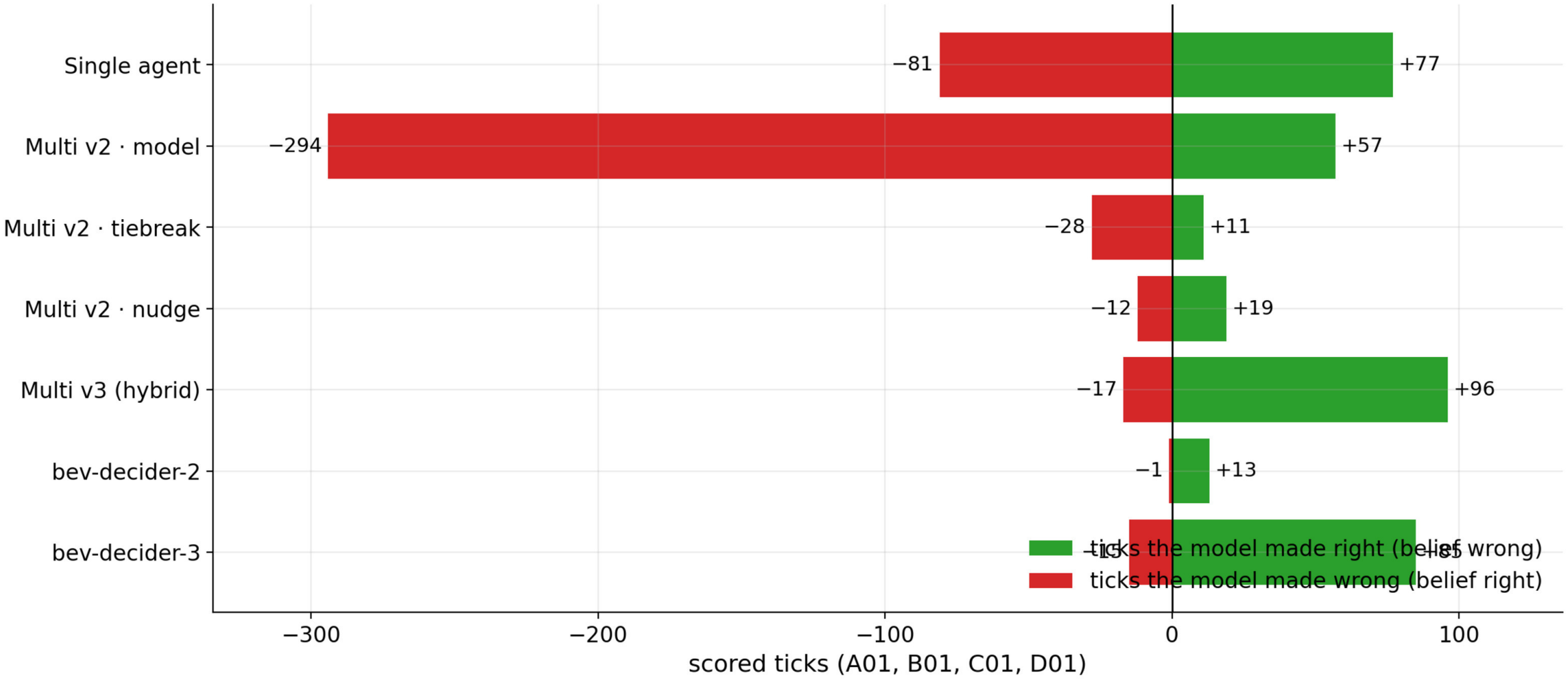
Higher is better for all five



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Where the model changed the answer, compared with belief

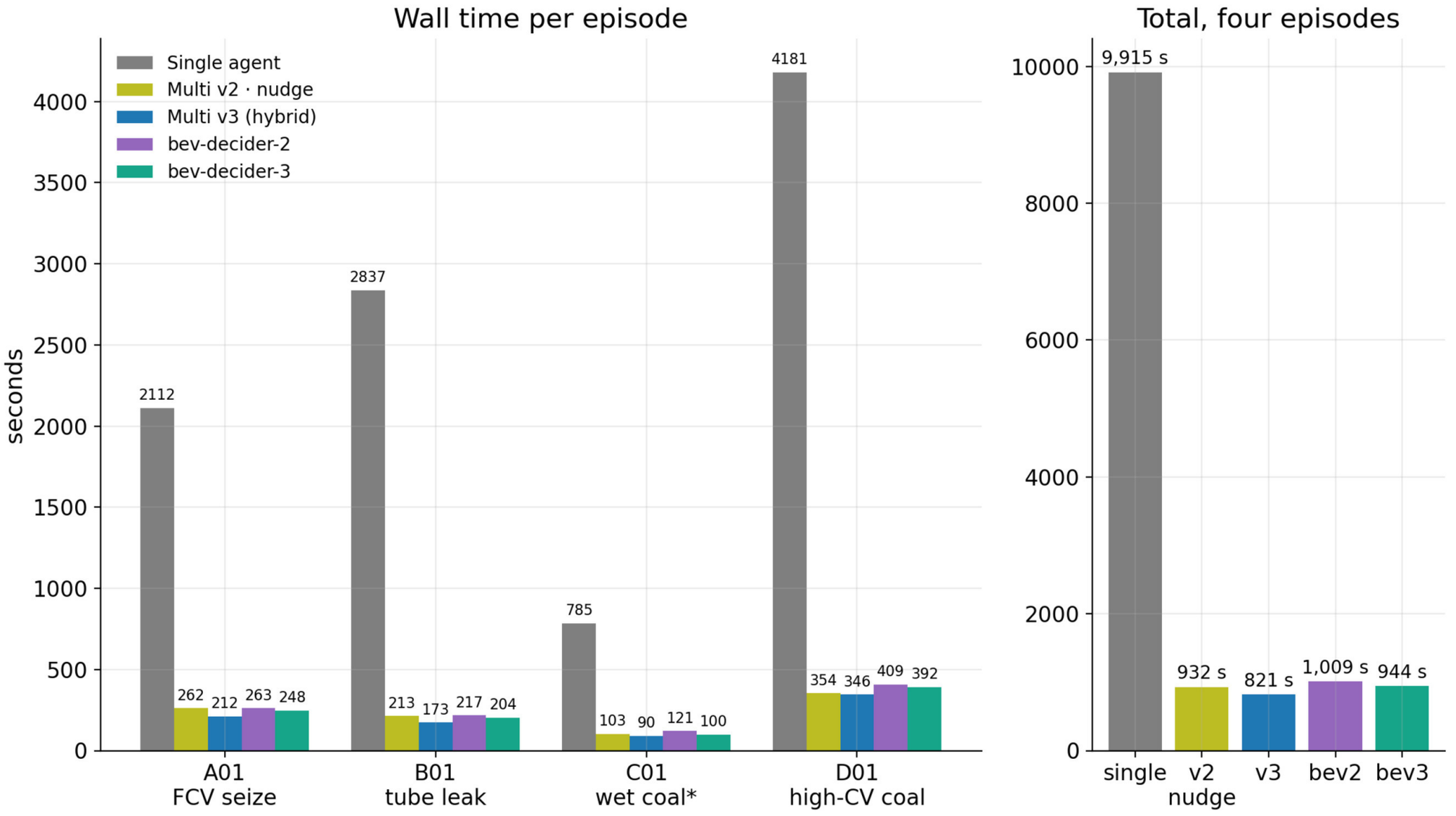
Net value of the language model = green minus red



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Speed: time to run the same four episodes

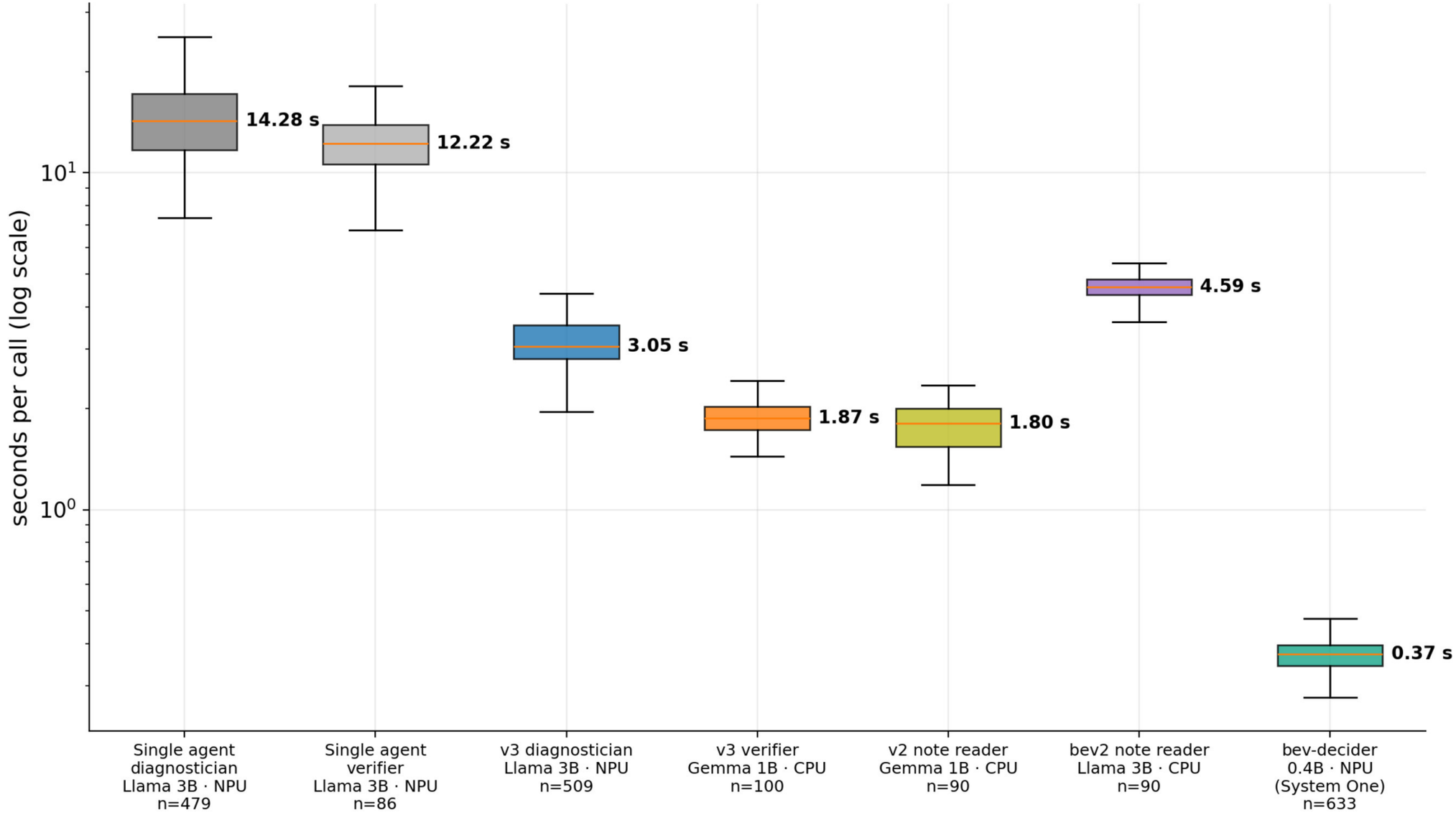
Lockstep: each tick waits for its own model calls. Single agent 9,915 s against v3 821 s (12.1× faster)



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Model call time by agent, lane and model

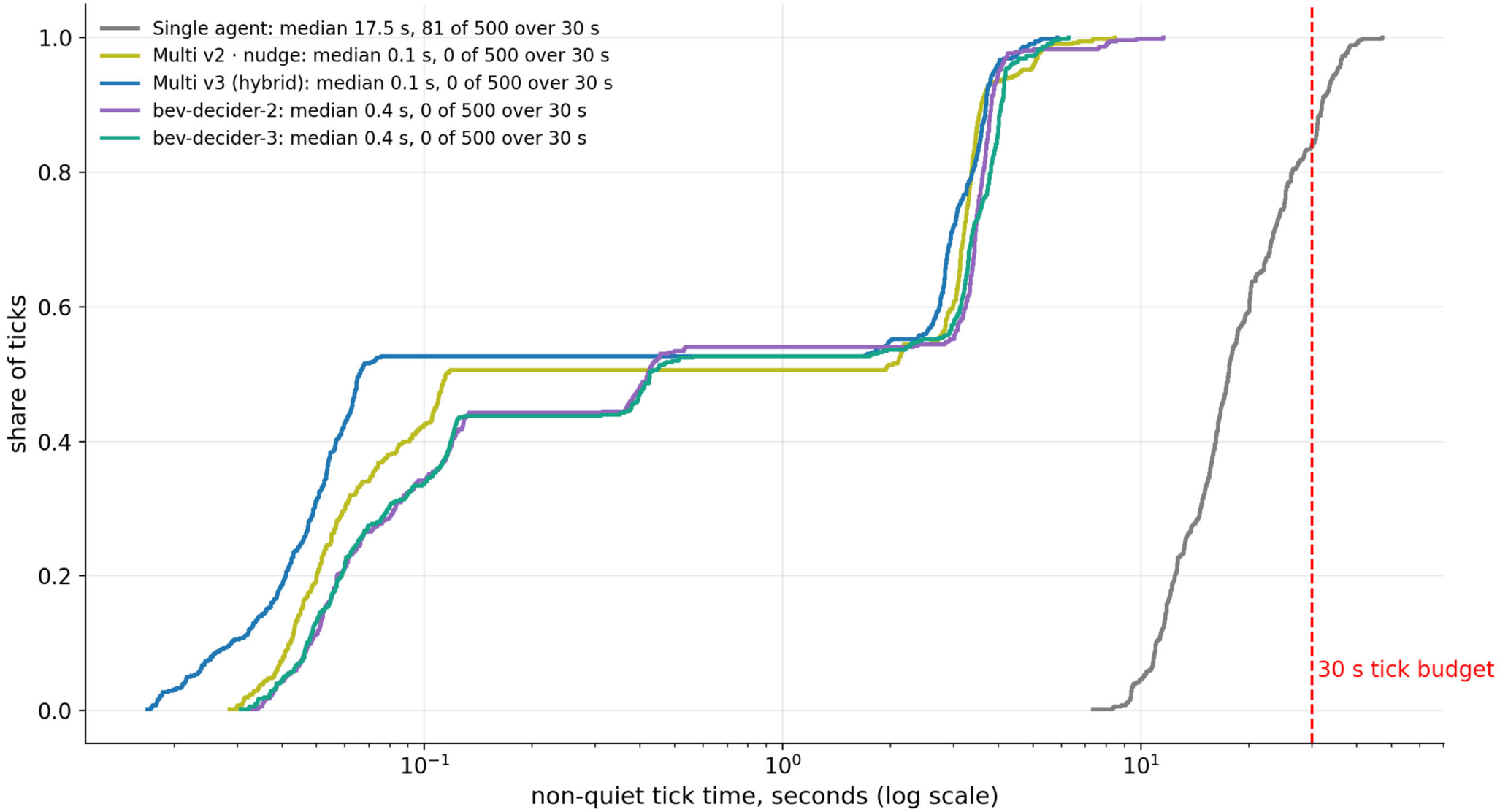
Board-side call time (box = middle 50%, whiskers 1.5 IQR, outliers hidden) · label = median



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Tick time: how long each non-quiet tick took

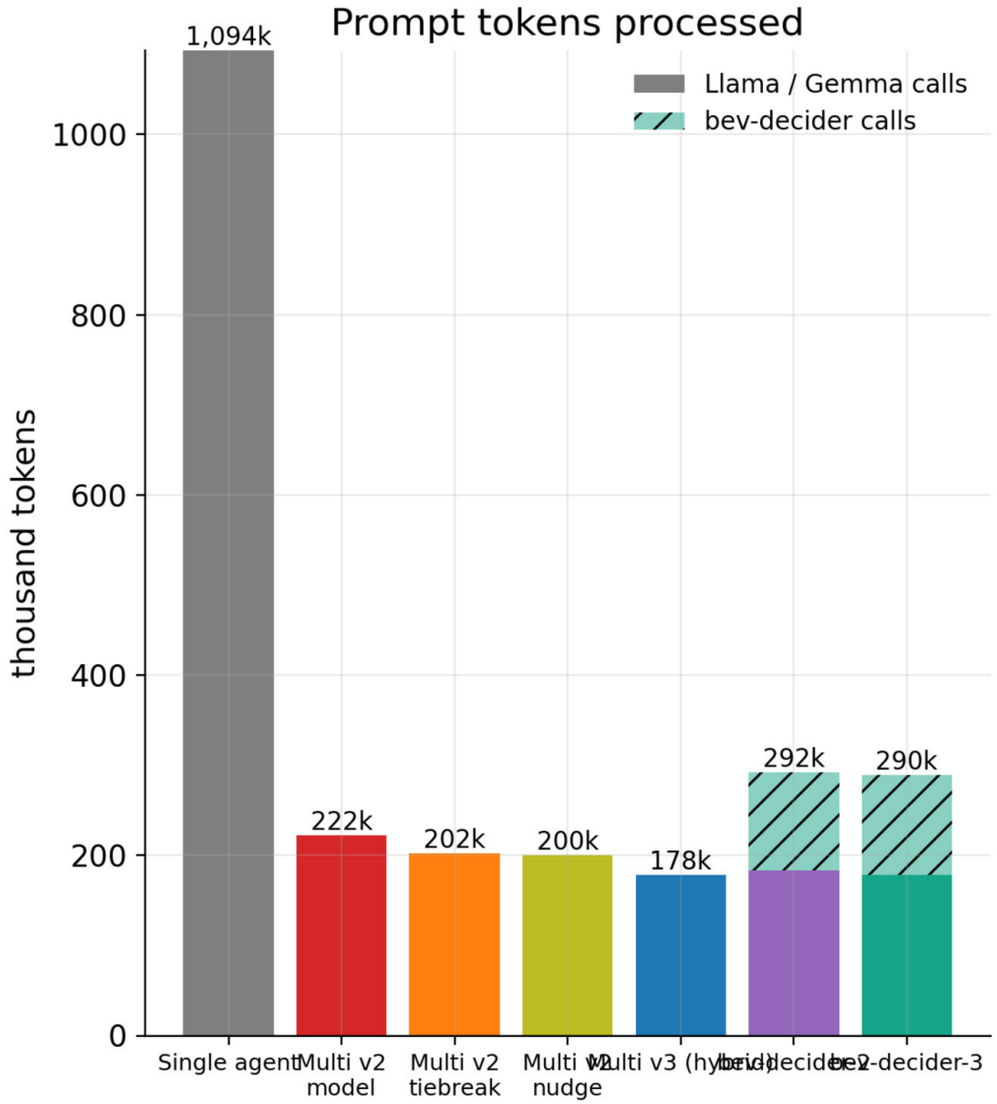
A01, B01, C01, D01 · cumulative distribution · tick time = laptop code + board model calls

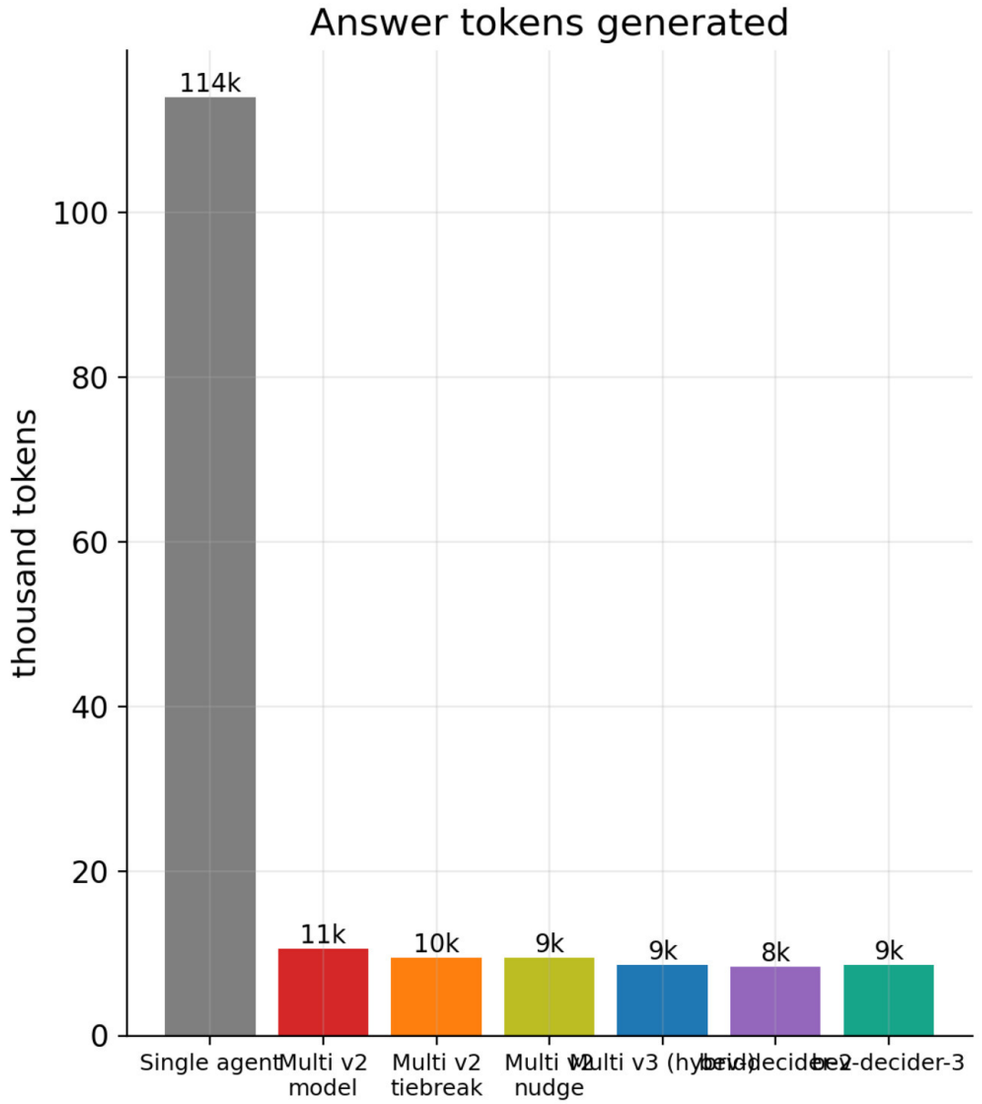


QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Tokens processed on the board, four fault episodes

Decode (answer) tokens are the expensive ones: ~14–16 tok/s on the NPU against 670–990 tok/s prefill



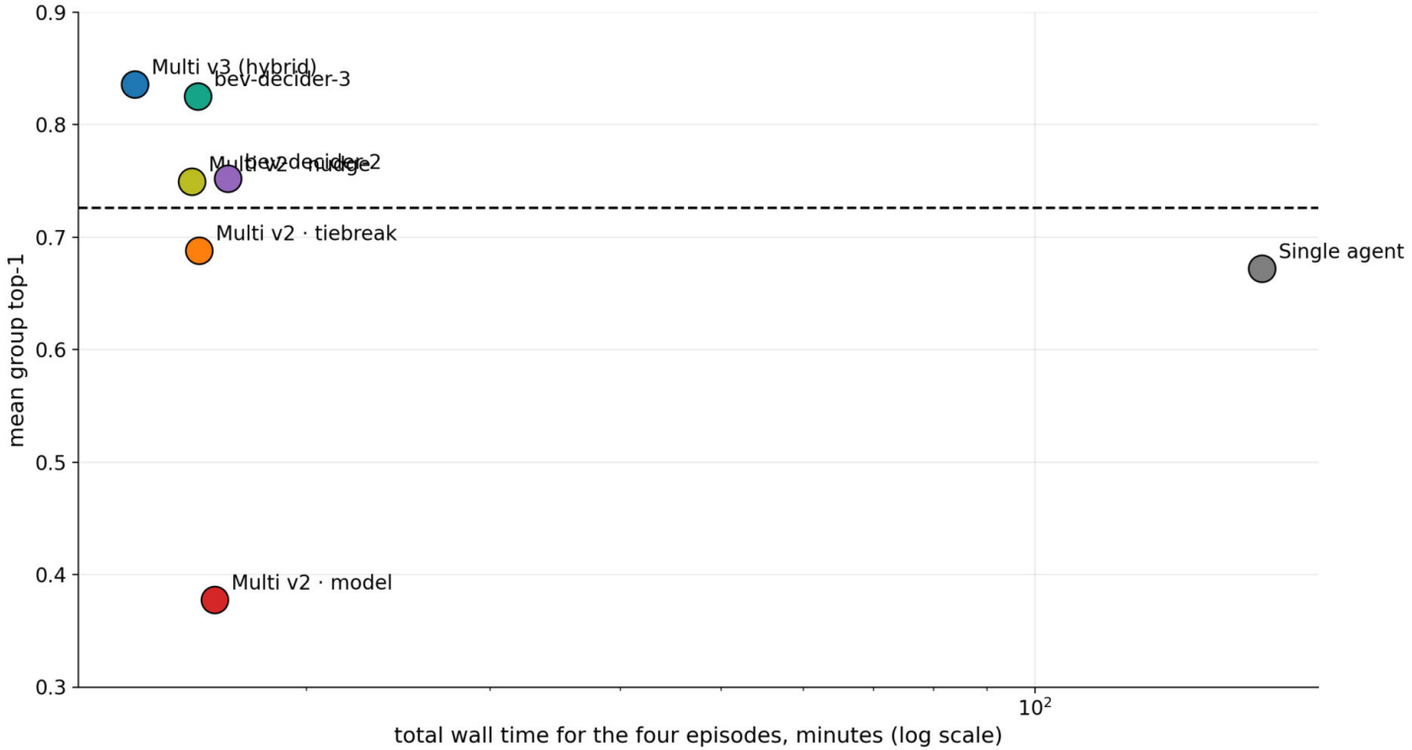


QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

belief alone 0.726

Accuracy against time: up and to the left is better

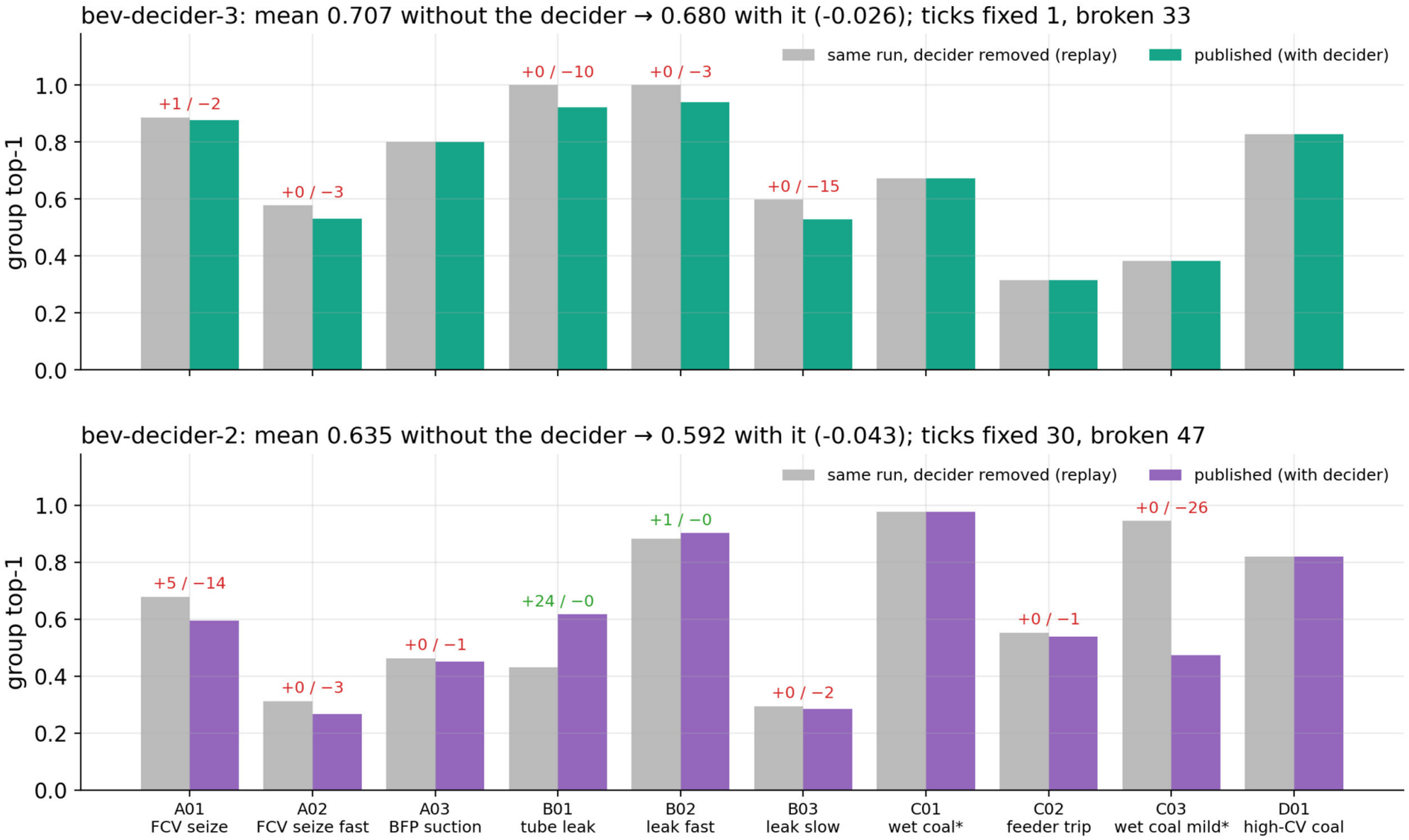
A01, B01, C01, D01



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

What bev-decider changed, measured inside the same runs

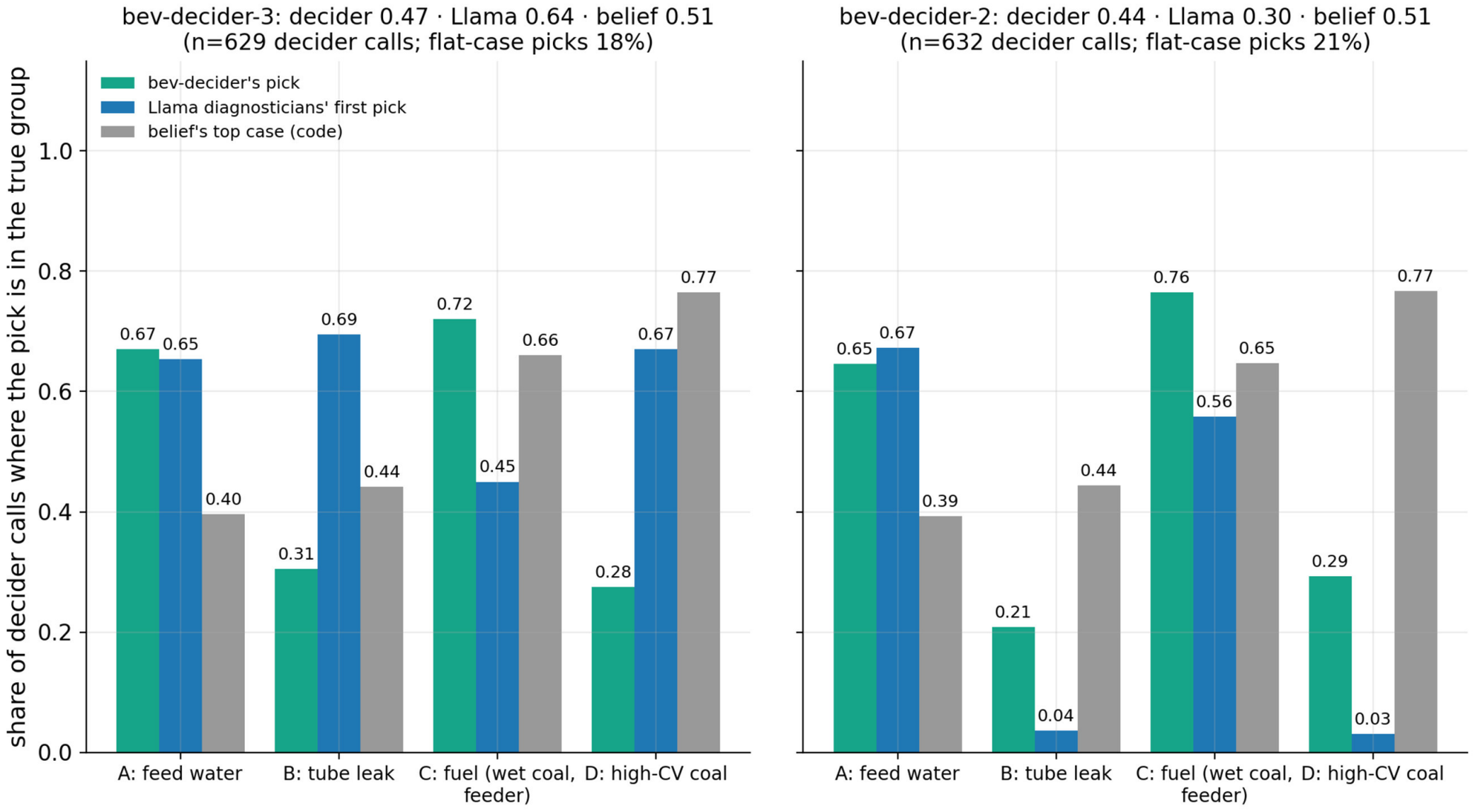
Each bev run replayed from its own saved answers with the decider's offsets removed: no run-to-run model noise · labels = ticks fixed / broken



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

How often each source names the right cause

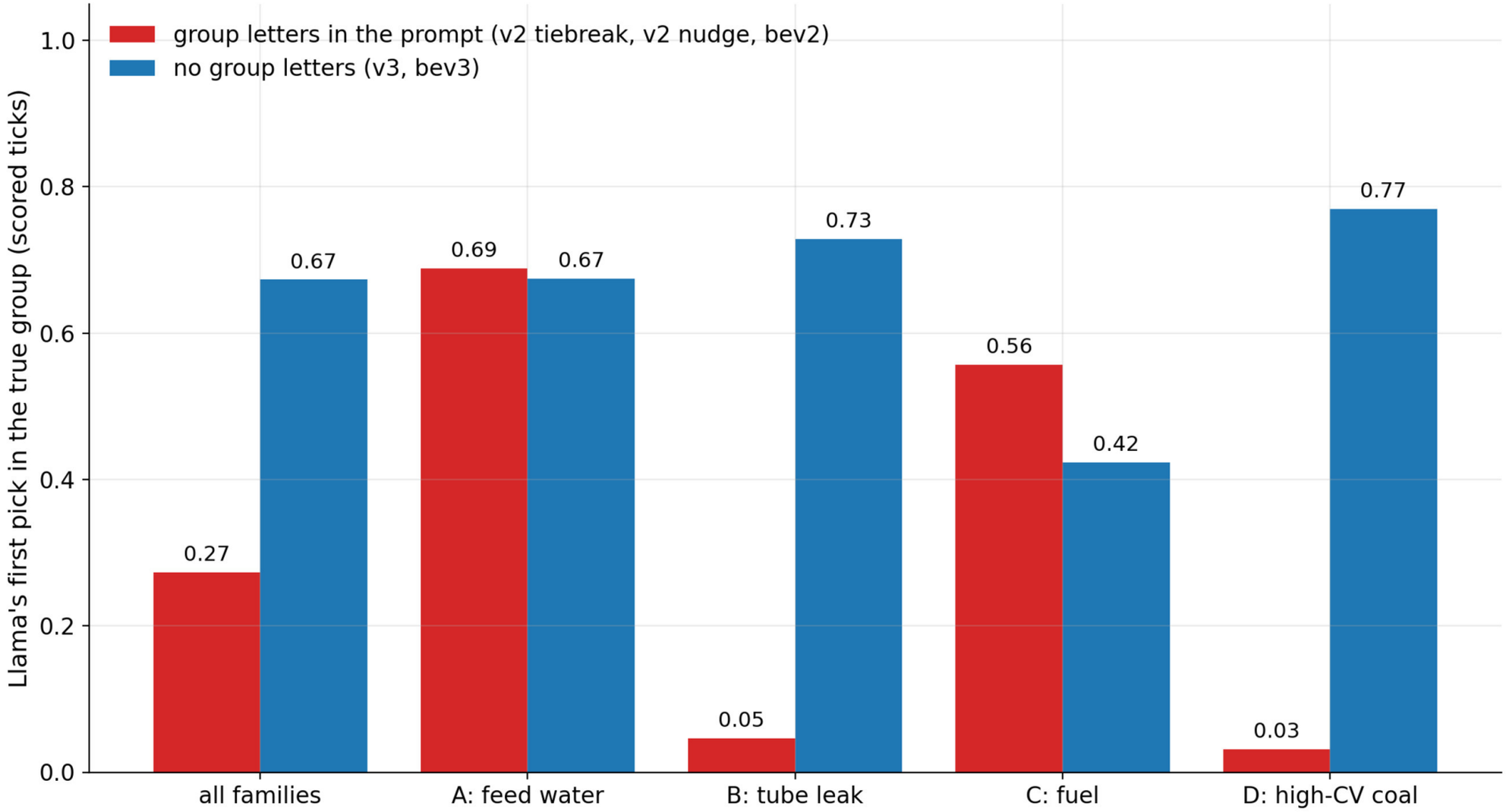
On the ticks where bev-decider was called (after fault onset), by fault family



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

A prompt shortcut: the group letter halved the model's accuracy

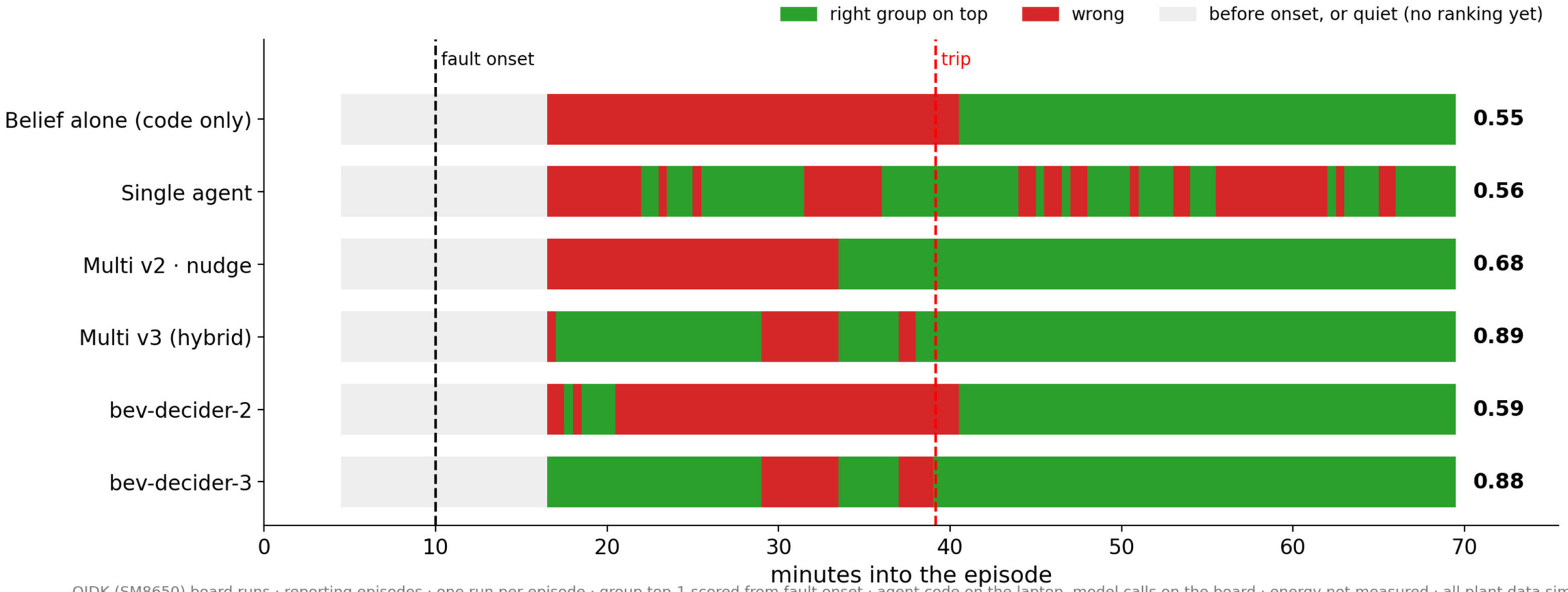
Same model (Llama 3.2 3B, NPU). With letters, the model tended to copy the most common letter instead of reading the facts · five independent board runs



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Tick by tick: A01 FCV seize

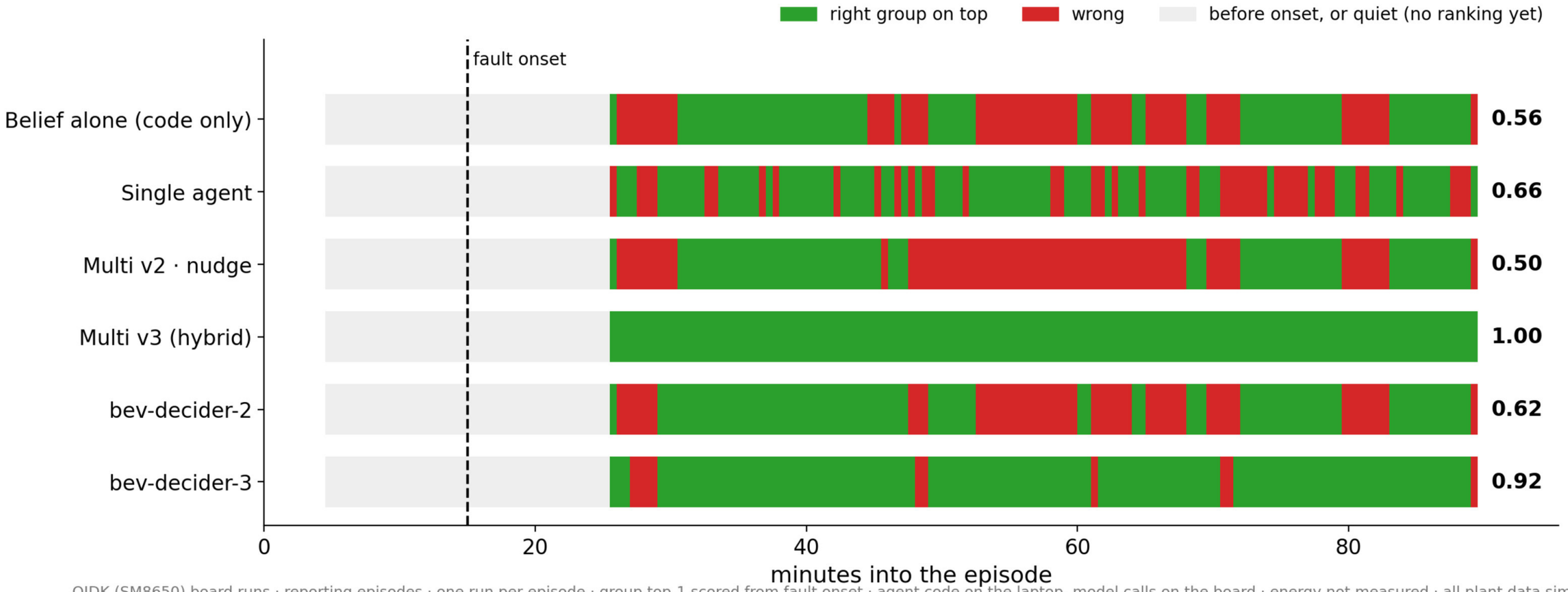
Each cell = one 30 s tick; the number on the right = group top-1 for the episode



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Tick by tick: B01 tube leak

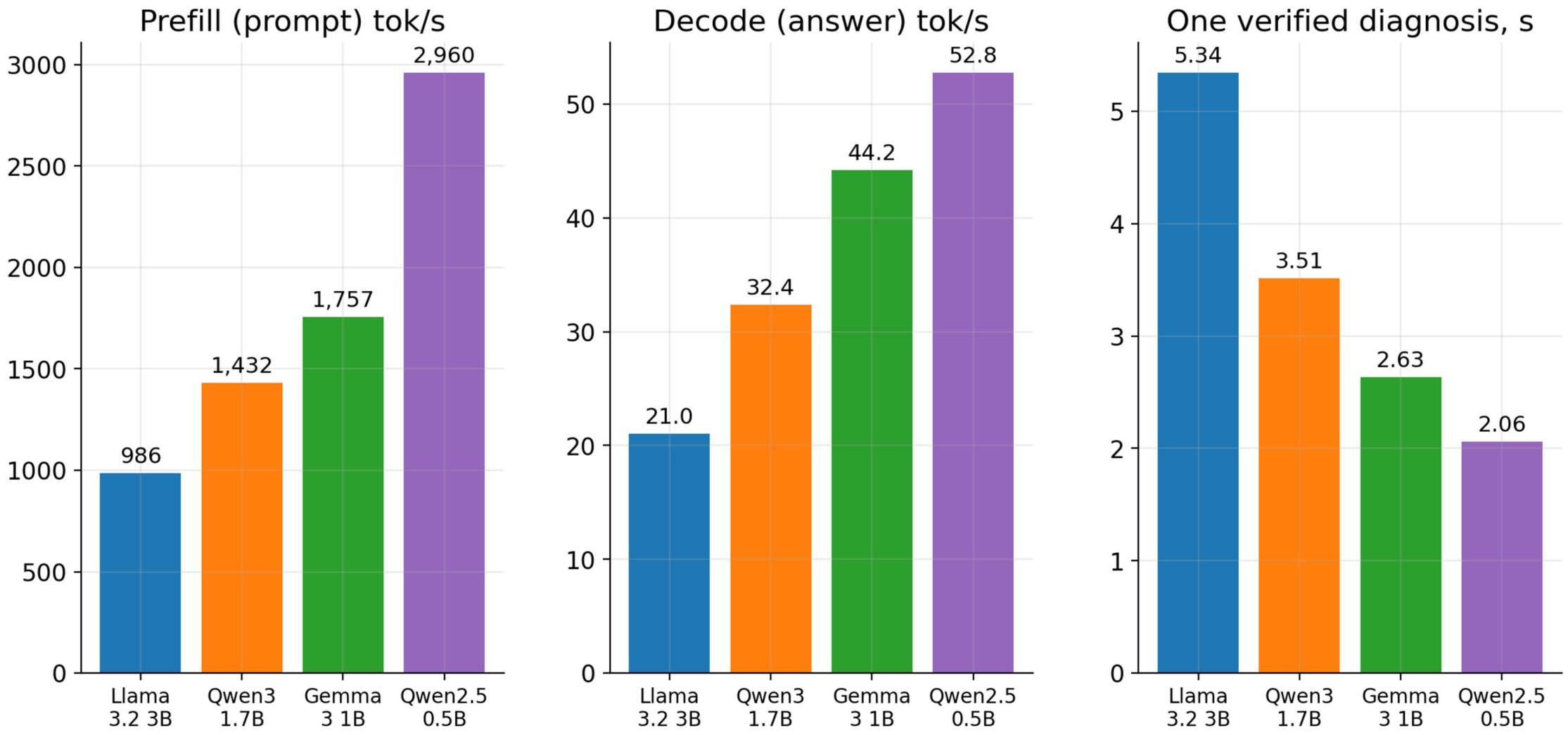
Each cell = one 30 s tick; the number on the right = group top-1 for the episode



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated

Four models on the Hexagon NPU (HTP0), llama.cpp b11371

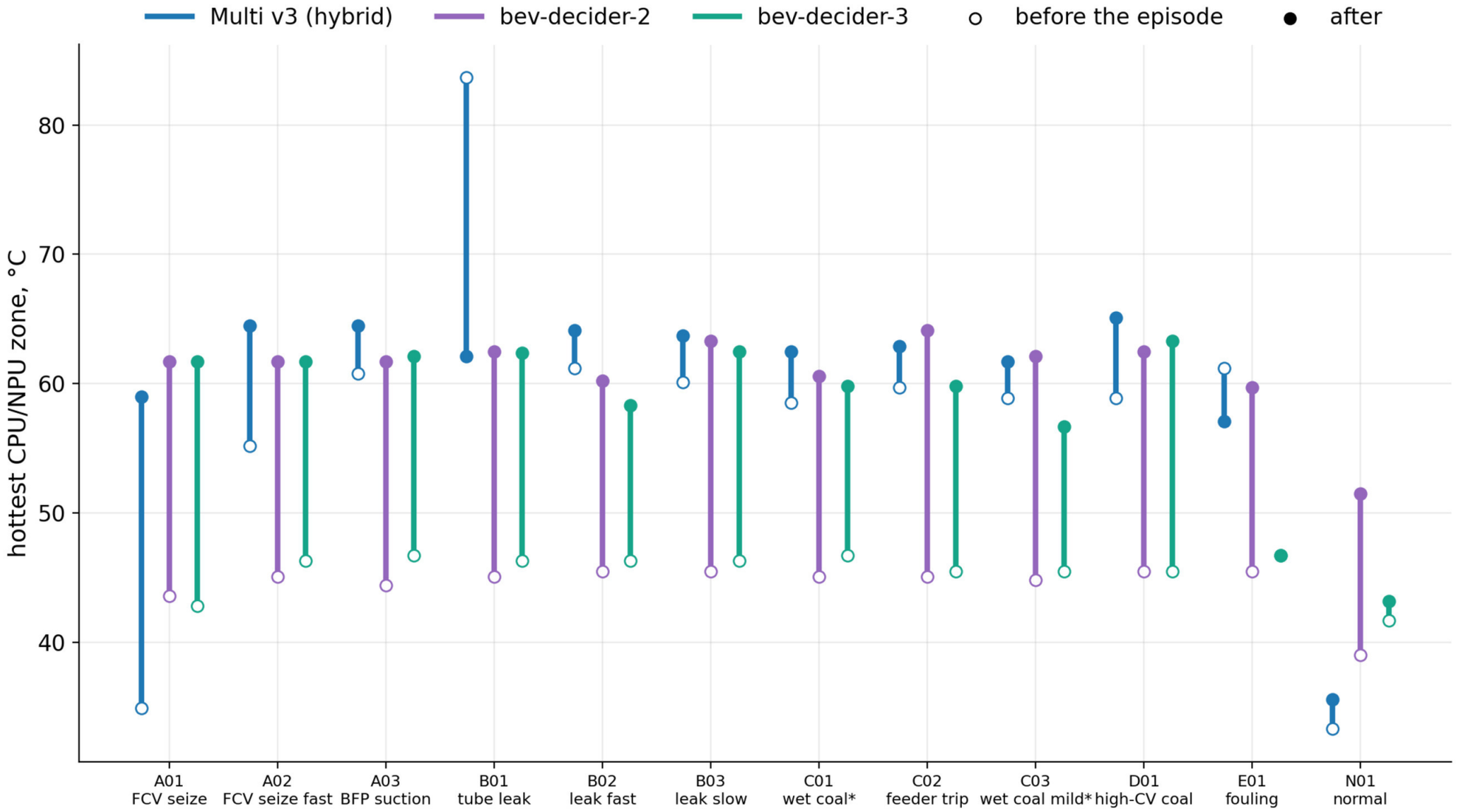
Pure Q4_0 weights, Q8_0 embedding, every layer on HTP0 · diagnosis = 700-token prompt + 60-token answer, plus a 350 + 30 verification



QIDK (SM8650) screening runs, 10 calls per model · llama-server -c 4096 -np 1 --cache-ram 0 · energy not measured

Chip temperature before and after each episode

The runner waits for the chip to cool before each episode · v3's episodes after its B01 restart started hot (cooling target taken from a hot reading)



QIDK (SM8650) board runs · reporting episodes · one run per episode · group top-1 scored from fault onset · agent code on the laptop, model calls on the board · energy not measured · all plant data simulated